

笔记前言：

本笔记的内容是去掉步骤的概述后，视频的所有内容。

本猴觉得，自己的步骤概述写的太啰嗦，大家自己做笔记时，应该每个人都有自己的最舒服最简练的写法，所以没给大家写。再是本猴觉得，不给大家写这个概述的话，大家会记忆的更深，掌握的更好！

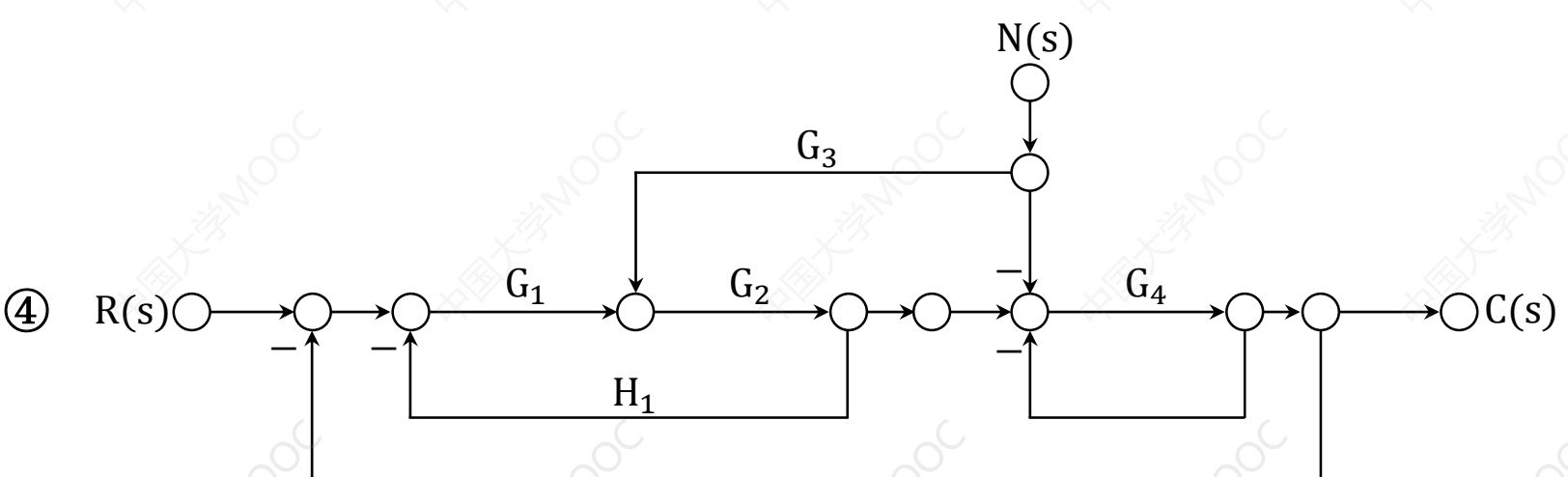
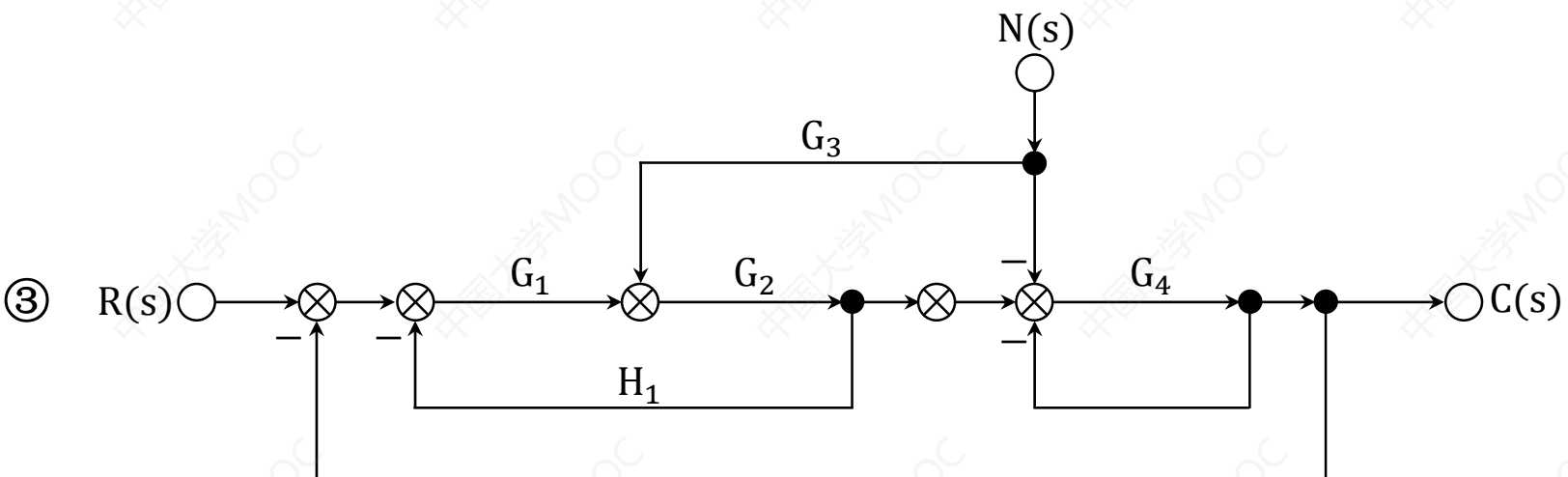
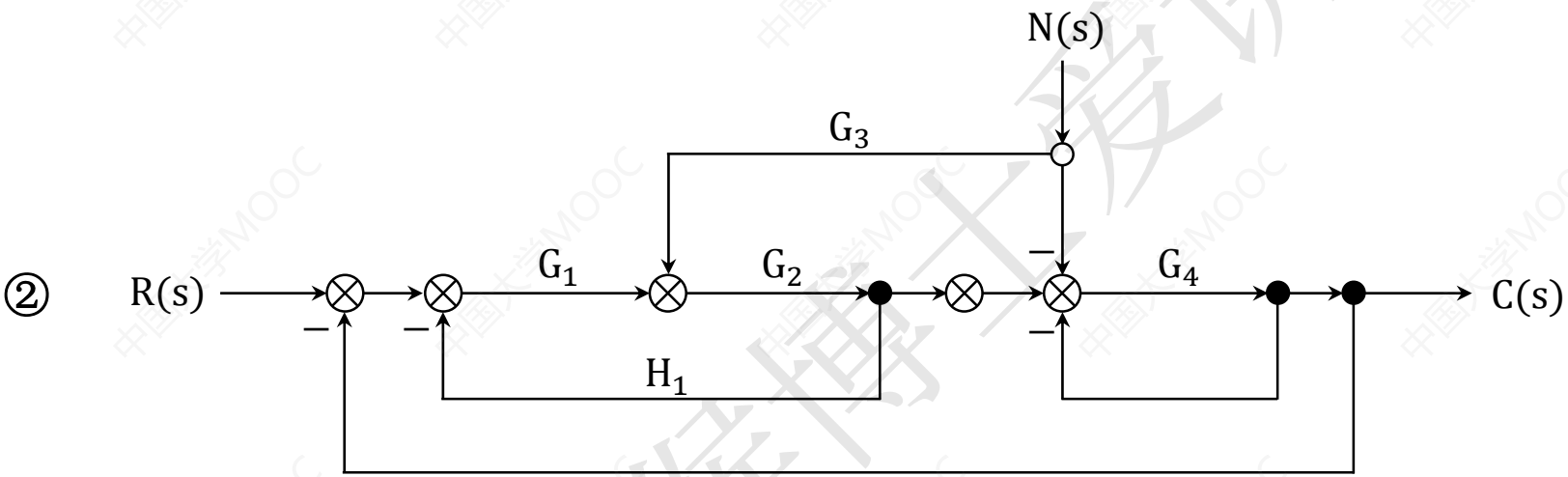
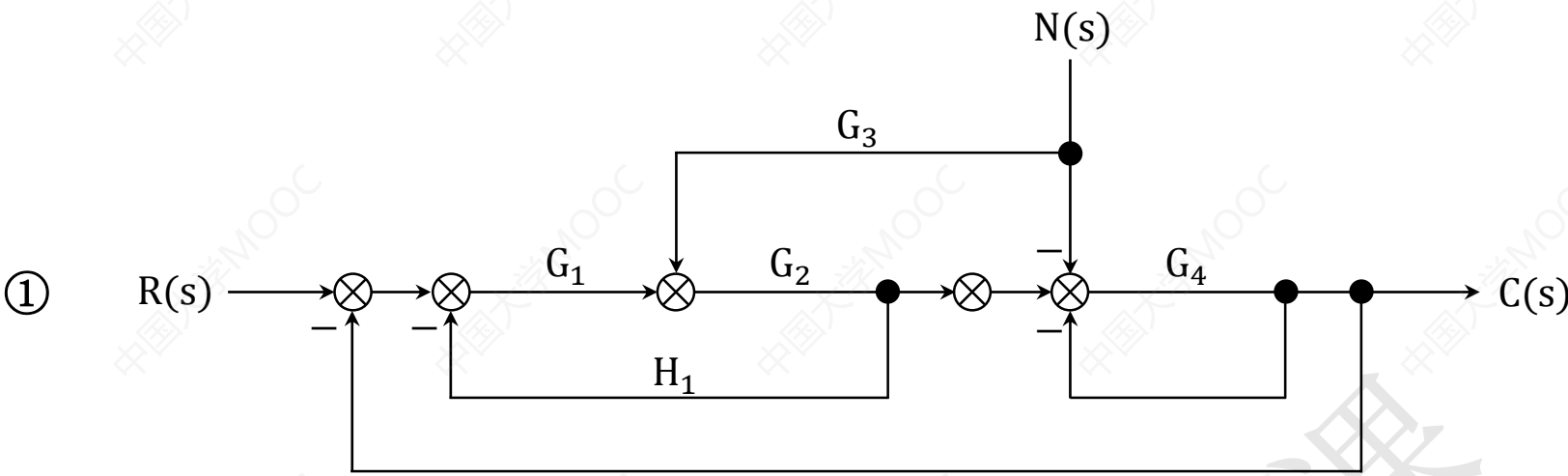
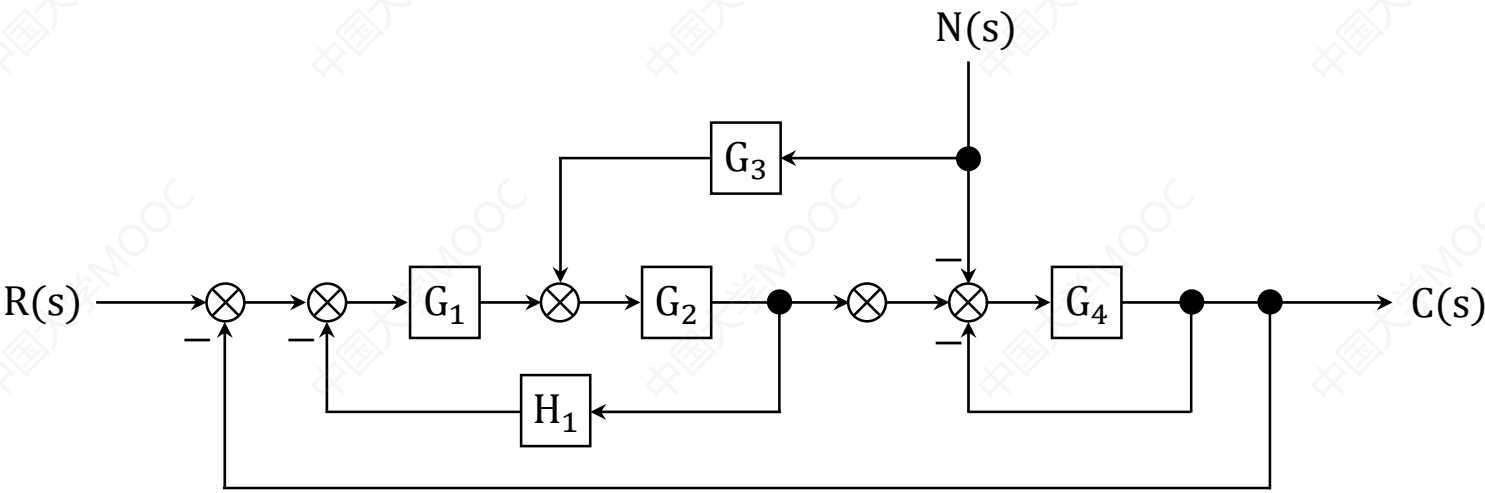
所以老铁！一定要过呀！不要辜负本猴的心意！~~~

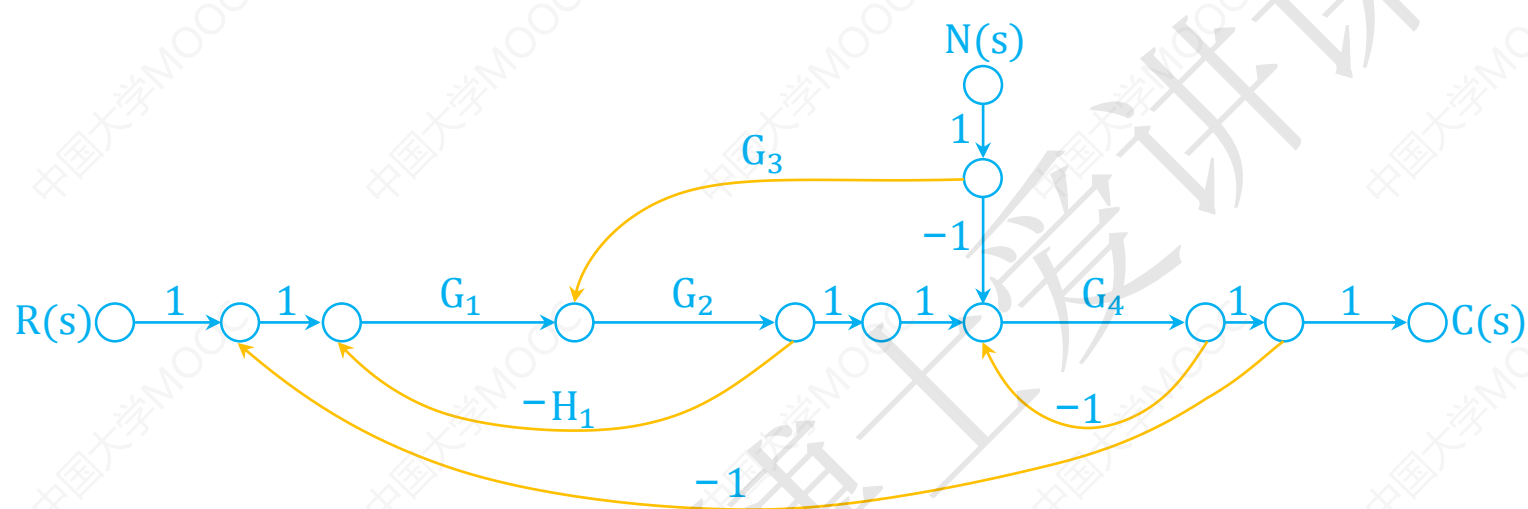
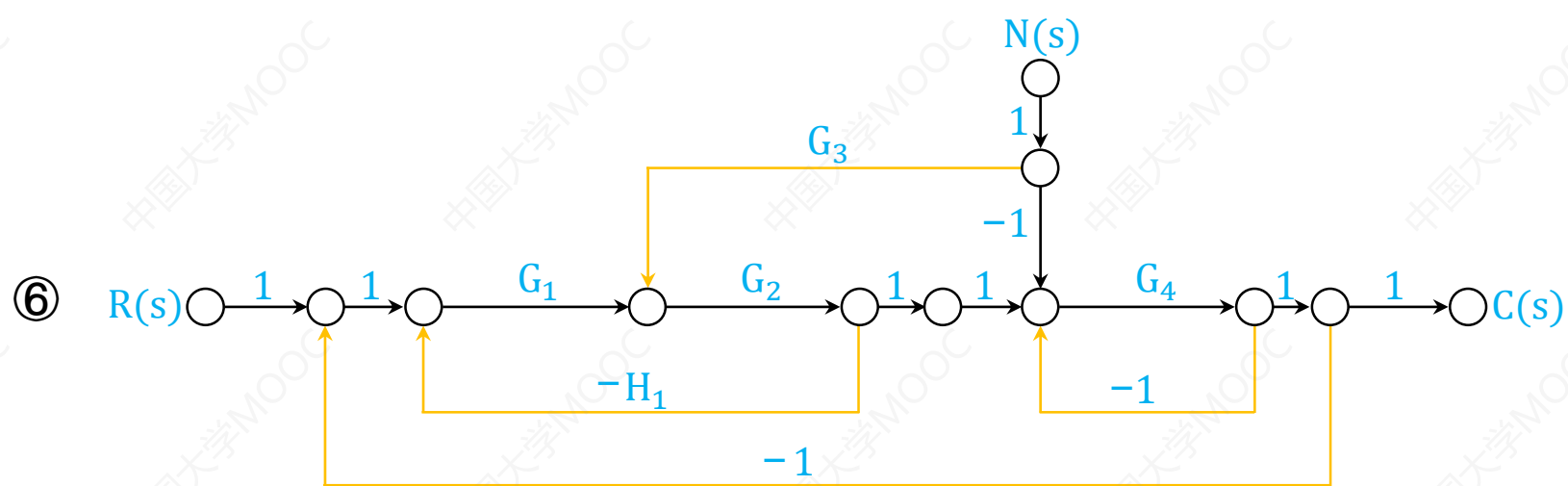
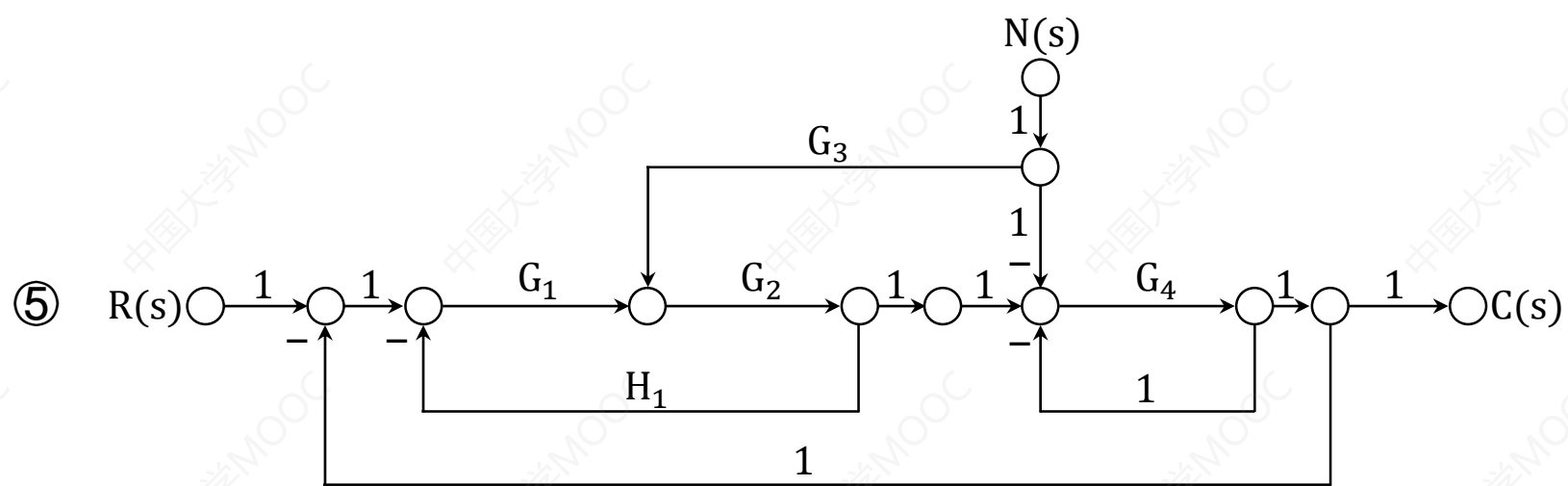
【祝逢考必过，心想事成~~~~】

【一定能过！！！！】

画信号流图

例. 如图是某系统的结构图，试画出系统的信号流图

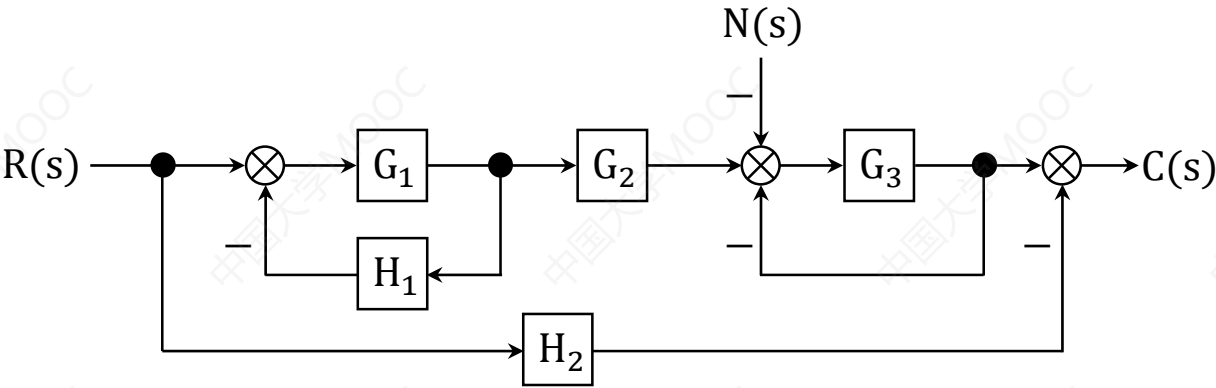




梅逊法求传递函数

例1. 如图是某系统的结构图，求传递函数 $\frac{C(s)}{R(s)}$

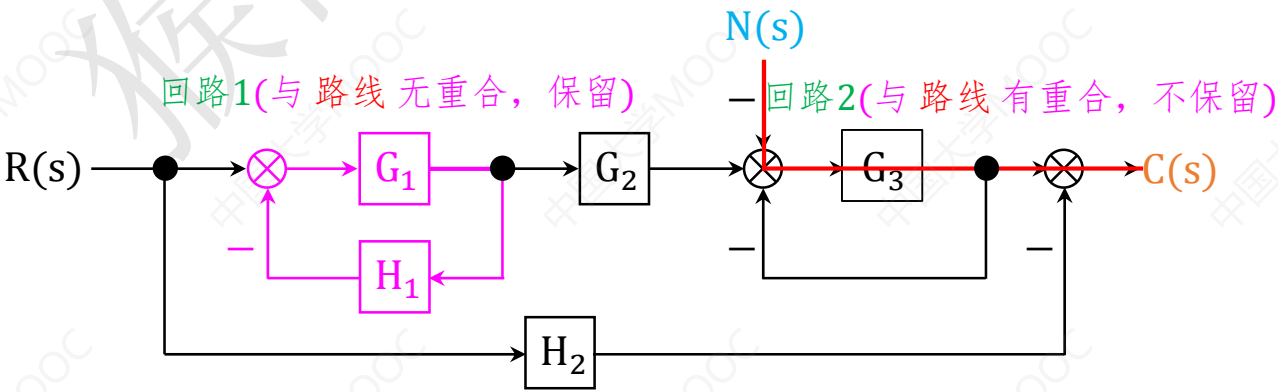
求 $\frac{C(s)}{R(s)}$: $\frac{C(s)}{R(s)} = \frac{\text{每个} p \cdot (1-b) \text{结果之和}}{1-L_{\text{和}}+a}$
求 $\frac{C(s)}{N(s)}$: 将上述步骤里所有的 $R(s)$ 改成 $N(s)$ 即可



路线1:
② $p = -H_2$
③ $b = -G_1H_1 - G_3$
④ $p \cdot (1 - b)$
 $= -H_2 - G_1H_1H_2 - G_3H_2$

路线2:
② $p = G_1G_2G_3$
③ $b = 0$
④ $p \cdot (1 - b) = G_1G_2G_3 \cdot (1 - 0) = G_1G_2G_3$
⑥ $L_{\text{和}} = -G_1H_1 + -G_3$
⑦ $a = G_1H_1G_3$
⑧ $\frac{C(s)}{R(s)} = \frac{\text{每个} p \cdot (1-b) \text{结果之和}}{1-L_{\text{和}}+a}$
 $= \frac{-H_2-G_1H_1H_2-G_3H_2+G_1G_2G_3}{1+G_1H_1+G_3+G_1H_1G_3}$

例1（改）. 如图是某系统的结构图，求传递函数 $\frac{C(s)}{N(s)}$



② $p = -G_3$
③ $b = -G_1H_1$
④ $p \cdot (1 - b)$
 $= -G_3 \cdot [1 - (-G_1H_1)]$
 $= -G_3 - G_1H_1G_3$
⑤ 找不到新路线，进行⑥
⑥ $L_{\text{和}} = -G_1H_1 + (-G_3)$
 $= -G_1H_1 - G_3$

⑦ $a = -(-G_1H_1G_3) = G_1H_1G_3$
⑧ $\frac{C(s)}{N(s)} = \frac{\text{每个} p \cdot (1-b) \text{结果之和}}{1-L_{\text{和}}+a}$
 $= \frac{-G_3-G_1H_1G_3}{1+G_1H_1+G_3+G_1H_1G_3}$
只有一条路线，对应只有一个 $p \cdot (1 - b)$

求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$

例1. 已知单位反馈系统的开环传递函数 $G(s) = \frac{K}{Ks^2+s+s^3} (K \neq 0)$,

求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{K}{Ks^2+s+s^3+K}$$
$$G(s)H(s) = G(s) = \frac{K}{Ks^2+s+s^3}$$

已知单位反馈系统的开环传递函数 $G(s)$:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)},$$
$$G(s)H(s) = G(s)$$

例2. 已知单位反馈系统的开环传递函数 $G(s) = \frac{1}{s(s+0.56)}$,

求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{1}{s^2+0.56s+1}$$
$$G(s)H(s) = G(s) = \frac{1}{s(s+0.56)}$$

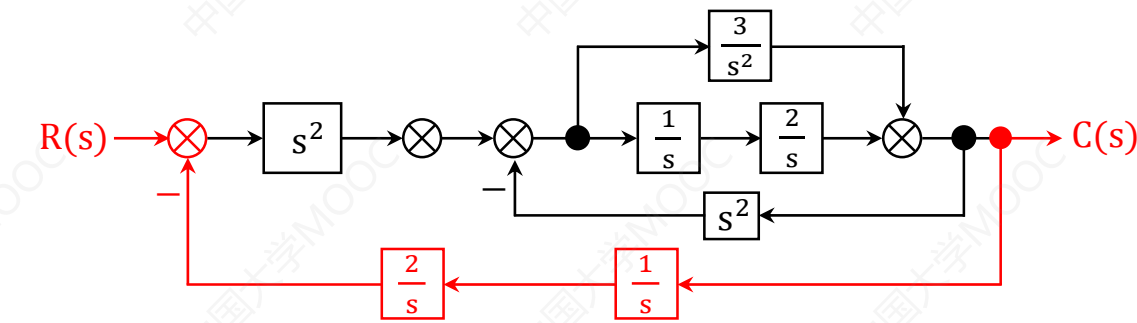
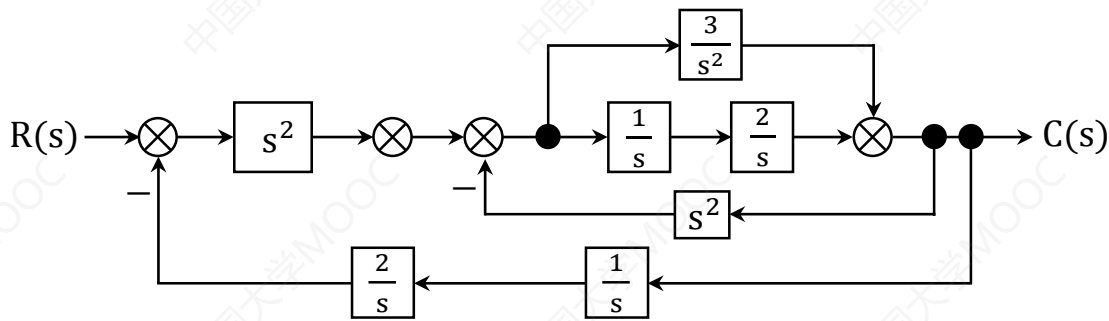
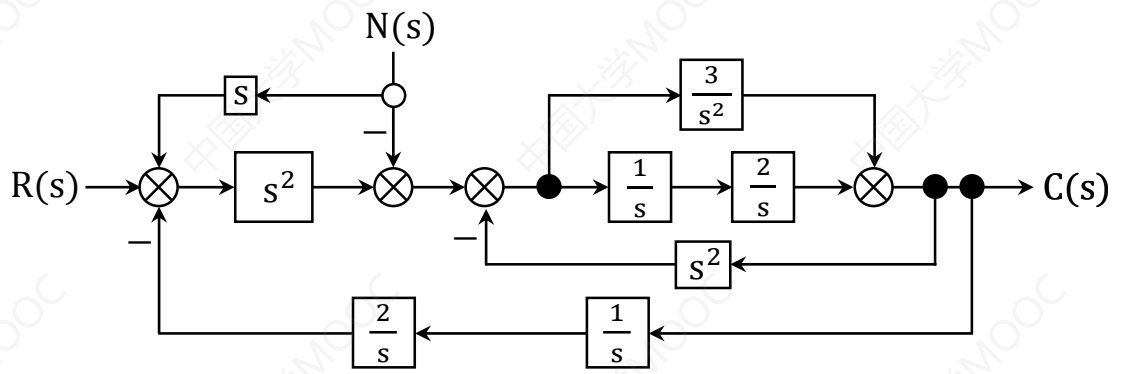
例3. 已知单位反馈系统的开环传递函数 $G(s) = \frac{K_1}{s(s+10)(0.2s+1)}$,

求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$

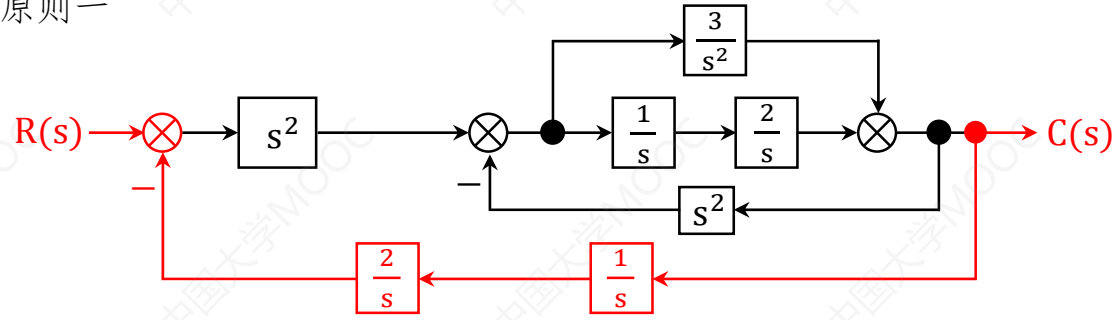
$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{K_1}{s(s+10)(0.2s+1)+K_1}$$
$$G(s)H(s) = G(s) = \frac{K_1}{s(s+10)(0.2s+1)}$$

例4. 如图是某系统的结构图，求闭环传递函数 $\frac{C(s)}{R(s)}$

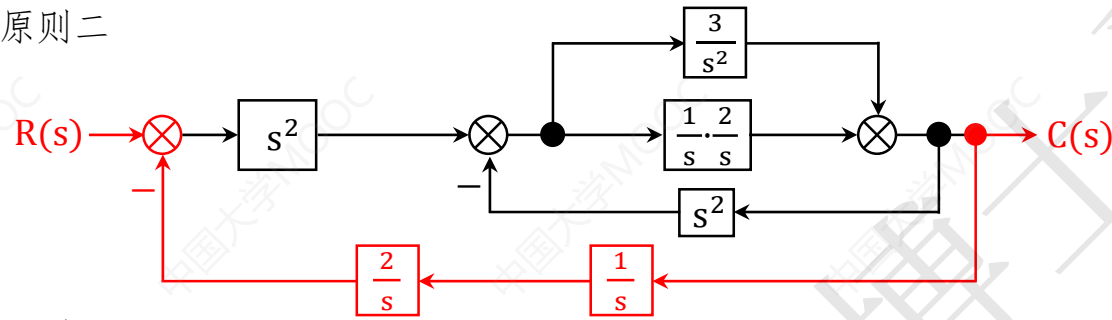
和开环传递函数 $G(s)H(s)$



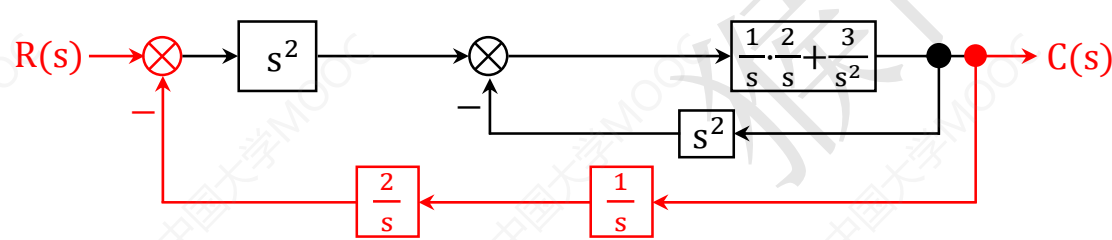
原则一



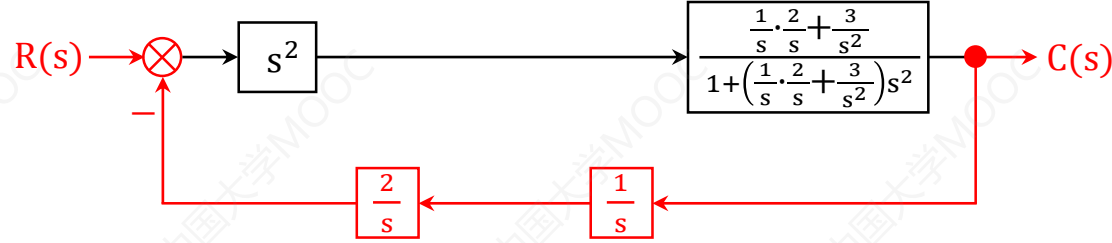
原则二



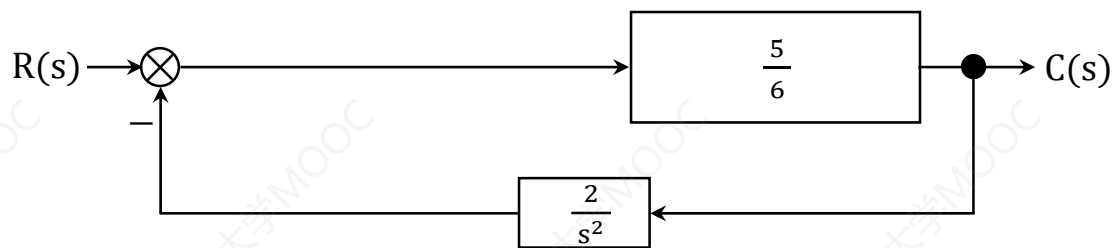
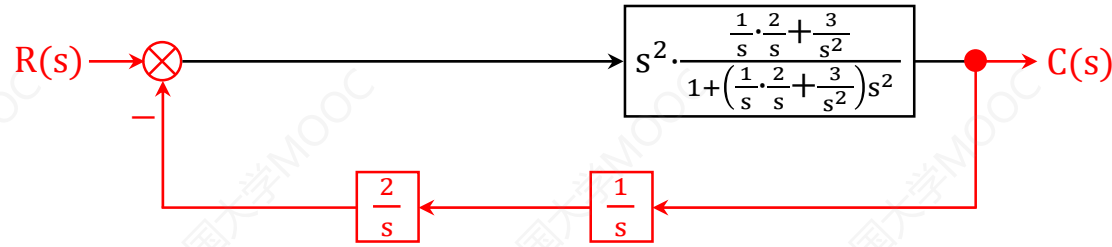
原则三



原则四



原则二



$$\frac{C(s)}{R(s)} = \frac{A}{1 \mp A \cdot B} = \frac{5s^2}{6s^2 + 10}$$

$$G(s)H(s) = A \cdot B = \frac{5}{3s^2}$$

已知结构图：

$$\frac{C(s)}{R(s)} = \frac{A}{1 \mp A \cdot B}; G(s)H(s) = A \cdot B$$

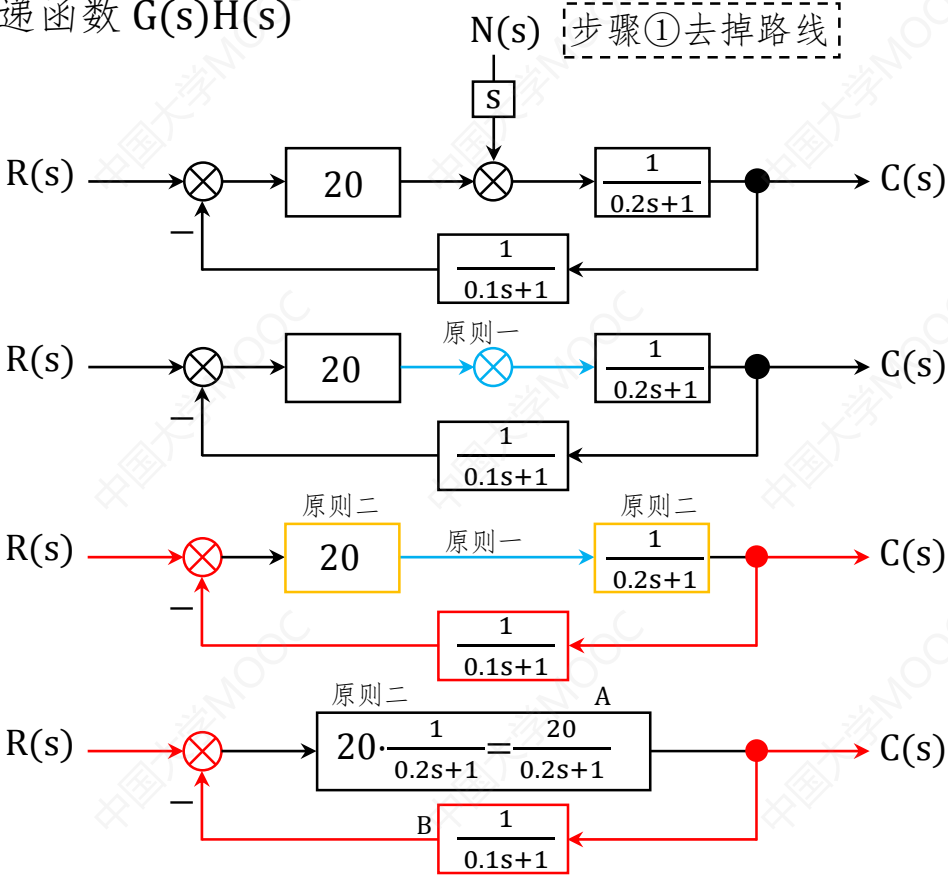
原则一： $\rightarrow \otimes \rightarrow$ 可变成 $\rightarrow$

原则二： $\rightarrow [a] \rightarrow [b]$ 可变成 $\rightarrow [a \cdot b]$

原则三： $\rightarrow [a] \rightarrow \otimes \rightarrow [b] \rightarrow$ 可变成 $\rightarrow [a \pm b]$

原则四： $\rightarrow \otimes \rightarrow [a] \rightarrow [b] \rightarrow$ 可变成 $\rightarrow \frac{a}{1 \mp ab}$

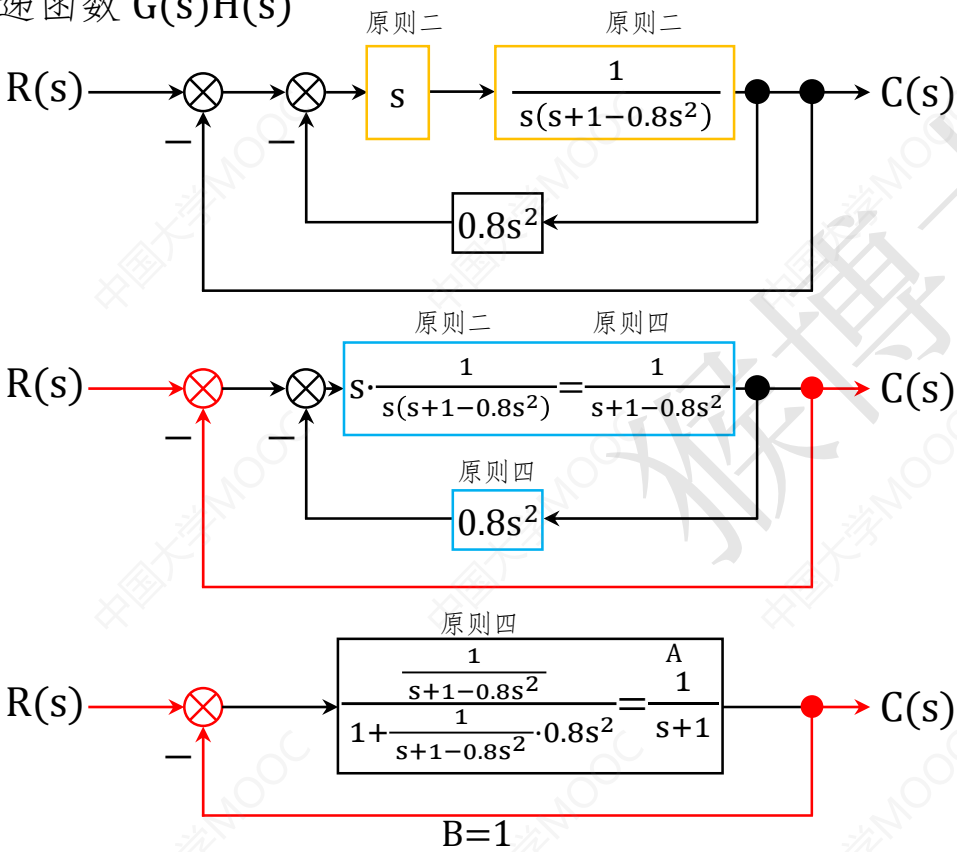
例5. 如图是某系统的结构图，求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$



$$\frac{C(s)}{R(s)} = \frac{A}{1+A \cdot B} = \frac{20(0.1s+1)}{(0.2s+1)(0.1s+1)+20}$$

$$G(s)H(s) = A \cdot B = \frac{20}{(0.2s+1)(0.1s+1)}$$

例6. 如图是某系统的结构图，求闭环传递函数 $\frac{C(s)}{R(s)}$ 和开环传递函数 $G(s)H(s)$



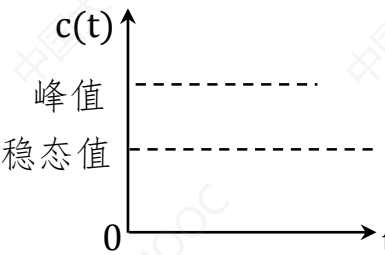
$$\frac{C(s)}{R(s)} = \frac{A}{1+A \cdot B} = \frac{1}{s+2}$$

$$G(s)H(s) = A \cdot B = \frac{1}{s+1}$$

二阶系统欠阻尼动态分析

名称	结果
衰减系数 σ	$\xi\omega_n$
阻尼振荡频率 ω_d	$\omega_n\sqrt{1-\xi^2}$
阻尼角 β	$\frac{\pi}{180^\circ}\arccos\xi$
上升时间 t_r	$\frac{\pi-\beta}{\omega_d}$
峰值时间 t_p	$\frac{\pi}{\omega_d}$
调节时间 t_s	$\Delta = 0.05$ 时, $\frac{3.5}{\sigma}$
	$\Delta = 0.02$ 时, $\frac{4.4}{\sigma}$
超调量 $\sigma_p\%$	$e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}}\times 100\%$

输入信号为单位阶跃信号时：

名称	结果
稳态值	$\lim_{s\rightarrow 0}\frac{C(s)}{R(s)}$
峰值	$(\sigma_p\%+1)\cdot \text{稳态值}$
概略响应曲线	a、画出下图  b、原点出发到峰值 c、降落至低于稳态值 d、在稳态值附近上下波动，最后回归到稳态值 e、最终点向下作虚线，与t轴交点为t_s f、给峰值、稳态值、t_s填上具体值

例1. 某单位反馈系统的开环传递函数 $G(s) = \frac{1}{s(s+0.56)}$ ，
求衰减系数 σ 、阻尼振荡频率 ω_d 、阻尼角 β 、上升时间 t_r 、
峰值时间 t_p 、 $\Delta = 0.05$ 时的调节时间 t_s 、超调量 $\sigma_p\%$ ，
输入信号为单位阶跃信号时的稳态值、峰值、概略响应曲线。

① $\frac{C(s)}{R(s)} = \frac{1}{s^2+0.56s+1}$ [第一章第3课学过]

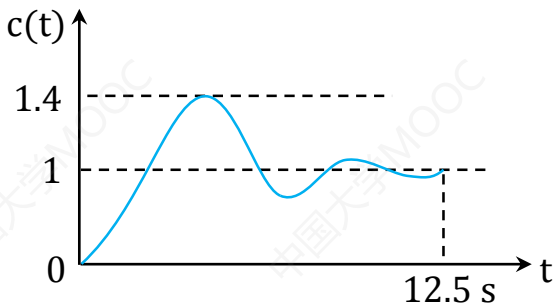
② 化简 $\frac{C(s)}{R(s)}$ ，得到a、b，求出 ω_n 、 ξ

a = 0.56, b = 1
 $\omega_n = \sqrt{b} = 1$ $\xi = \frac{a}{2\sqrt{b}} = 0.28$

③ $\sigma = 0.28$ $\omega_d = 0.96 \text{ rad/s}$ $\beta = 1.29 \text{ rad}$ $t_r = 1.93 \text{ s}$
 $t_p = 3.27 \text{ s}$ $t_s = 12.5 \text{ s}$ $\sigma_p\% = 40\%$

稳态值: $\lim_{s\rightarrow 0}\frac{C(s)}{R(s)} = \lim_{s\rightarrow 0}\frac{1}{s^2+0.56s+1} = 1$

峰值: $(\sigma_p\%+1)\cdot \text{稳态值} = 1.4$



劳斯判据

例1. 已知系统的特征方程为 $2 - 3s + s^3 = 0$ ，试用
劳斯判据判断系统稳定性并判断正实部根的个数

特征式子 $= 2 - 3s + s^3$

s^3	1	-3
s^2	0	2
s^1		
s^0	2	

s^3	A: 1	C: -3
s^2	B: ϵ	D: 2
s^1	$\frac{BC-AD}{B} = \frac{\epsilon \cdot (-3) - 1 \cdot 2}{\epsilon} = \frac{-3\epsilon - 2}{\epsilon}$	
s^0	2	

s^3	A: 1	-3	C: 0
s^2	B: ϵ	2	D: 0
s^1	$\frac{-3\epsilon - 2}{\epsilon}$	$\frac{BC-AD}{B} = \frac{\epsilon \cdot 0 - 1 \cdot 0}{\epsilon} = 0$	
s^0	2		

s^3	1	正	-3	
s^2	ϵ	正	2	
s^1	$\frac{-3\epsilon - 2}{\epsilon}$	负	0	
s^0	2	正		
第一次切换 第二次切换 共切换2次				

系统不稳定，且 系统正实部根的个数=2

例2. 已知单位反馈系统的开环传递函数 $G(s) = \frac{K}{Ks^2 + s + s^3}$ ($K \neq 0$)
试用劳斯判据判断系统稳定性并判断正实部根的个数

$$\frac{C(s)}{R(s)} = \frac{K}{Ks^2 + s + s^3 + K}$$

特征式子 $= Ks^2 + s + s^3 + K$

s^3	A: 1	C: 1
s^2	B: K	D: K
s^1		
s^0	K	

s^3	A: 1	C: 1
s^2	B: K	D: K
s^1	$\frac{BC-AD}{B} = \frac{K \cdot 1 - 1 \cdot K}{K} = 0$	
s^0	K	

s^3	A: 1	1	C: 0
s^2	B: K	K	D: 0
s^1	0	$\frac{BC-AD}{B} = \frac{K \cdot 0 - 1 \cdot 0}{K} = 0$	
s^0	K		

$s^{\text{最高}}$	$s^{\text{最高的系数}}$	$s^{\text{最高}-2 \text{的系数}}$	$\cdots$	$s^0 \text{的系数(即常数项)}$
$s^{\text{最高}-1}$	$s^{\text{最高}-1 \text{的系数}}$	$s^{\text{最高}-3 \text{的系数}}$	$\cdots$	
$\vdots$				
s^0	常数项			

填行公式：

s^{a+2}	A				C	
s^{a+1}	B				D	
s^a			$\frac{BC-AD}{B}$			

s^3	1	1
s^2	K	K
s^1	上面的数 $\times$ [上面行表头 s 的指数 $+2-2 \times 1$]	上面的数 $\times$ [上面行表头 s 的指数 $+2-2 \times 2$]
s^0	K	

s^3	1	正	1
s^2	K	负	K
s^1	2K	负	0
s^0	K	负	
第一次切换 没切换 没切换 共切换1次			

当 $K > 0$ 时，
系统稳定，且 系统正实部根的个数=0
当 $K < 0$ 时，
系统不稳定，且 系统正实部根的个数=1

静态误差系数法求稳态误差

例1. 某单位反馈系统的开环传递函数 $G(s) = \frac{K_1}{s(s+10)(0.2s+1)}$
设输入信号是单位斜坡输入信号，若系统稳定，用静态误差系数法求输入信号作用下的稳态误差 $e_{ss}(\infty)$

① $G(s)H(s) = \frac{K_1}{s(s+10)(0.2s+1)}$
 $= \frac{K_1 \cdot s^0}{0.2s^3+3s^2+10s}$

第一章第3课学过

② $k_p = \infty$
 $k_v = \frac{K_1}{10}$
 $k_a = 0$

③ $e_{ss}(\infty) = \frac{1}{k_v}$
 $= \frac{10}{K_1}$

例2. 如图是某系统的结构图，设输入信号 $r(t) = 2 + 3t$ ，若系统稳定，用静态误差系数法求输入信号作用下的稳态误差 $e_{ss}(\infty)$

① $G(s)H(s) = \frac{20}{(0.2s+1)(0.1s+1)}$
 $= \frac{20 \cdot s^0}{0.02s^2+0.3s^1+1 \cdot s^0}$

第一章第3课学过

② $k_p = 20$
 $k_v = 0$
 $k_a = 0$

③ $e_{ss}(\infty) = \frac{2}{1+k_p} + \frac{3}{k_v}$ 【求极限时， $\frac{\text{非零数}}{0} = \infty$ 】
 $= \infty$ 【数 + $\infty = \infty$ 】

【 $K = \frac{\text{分子里 } s \text{ 的最小指数所在项前的系数}}{\text{分母里 } s \text{ 的最小指数所在项前的系数}}$ 】

分母里 s 的最小指数	k_p	k_v	k_a
0	K	0	0
1	∞	K	0
2	∞	∞	K
3	∞	∞	∞

若输入信号是汉字，则 $e_{ss}(\infty) = \begin{cases} \text{输入单位阶跃, } \frac{1}{1+k_p} \\ \text{输入单位斜坡, } \frac{1}{k_v} \\ \text{输入单位加速度, } \frac{1}{k_a} \end{cases}$

若输入信号是 $r(t)$ ，则 $e_{ss}(\infty) = r(t)$ 结果里的 $\begin{cases} \text{常数项 变成 } \frac{\text{常数}}{1+k_p} \\ \text{常数} \cdot t \text{ 项 变成 } \frac{\text{常数}}{k_v} \\ \text{常数} \cdot t^2 \text{ 项 变成 } \frac{2 \cdot \text{常数}}{k_a} \end{cases}$

若输入信号是 $R(s)$ ，则 $e_{ss}(\infty) = R(s)$ 结果里的 $\begin{cases} \frac{\text{常数}}{s} \text{ 项 变成 } \frac{\text{常数}}{1+k_p} \\ \frac{\text{常数}}{s^2} \text{ 项 变成 } \frac{\text{常数}}{k_v} \\ \frac{\text{常数}}{s^3} \text{ 项 变成 } \frac{\text{常数}}{k_a} \end{cases}$

终值定理法求稳态误差

例1. 某单位反馈系统的开环传递函数 $G(s) = \frac{K_1}{s(s+10)(0.2s+1)}$ ，设输入信号是单位斜坡输入信号，若系统稳定，用终值定理法求输入信号作用下的稳态误差 $e_{ss}(\infty)$

① $\frac{C(s)}{R(s)} = \frac{K_1}{s(s+10)(0.2s+1)+K_1}$ [第一章第3课学过]

② $H(s) = 1$

③ $R(s) = \frac{1}{s^2}$

④
$$\begin{aligned} e_{ss}(\infty) &= \lim_{s \rightarrow 0} s \cdot \left[1 - \frac{C(s)}{R(s)} H(s) \right] \cdot R(s) \\ &= \lim_{s \rightarrow 0} s \cdot \left[1 - \frac{K_1}{s(s+10)(0.2s+1)+K_1} \times 1 \right] \cdot \frac{1}{s^2} \\ &= \lim_{s \rightarrow 0} \frac{s(s+10)(0.2s+1)}{s(s+10)(0.2s+1)+K_1} \cdot \frac{1}{s^2} \\ &= \lim_{s \rightarrow 0} \frac{(s+10)(0.2s+1)}{s(s+10)(0.2s+1)+K_1} \quad \text{接近} \frac{(0+10) \times (0.2 \times 0 + 1)}{0 \times (0+10) \times (0.2 \times 0 + 1) + K_1} = \frac{10}{K_1} \\ &= \frac{10}{K_1} \end{aligned}$$

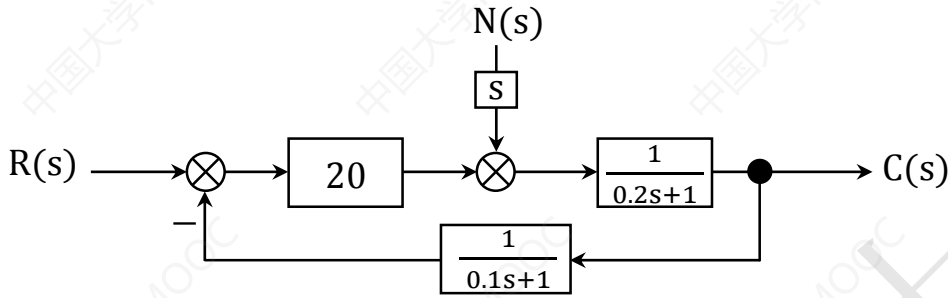
若输入信号是汉字，则 $R(s) = \begin{cases} \text{输入单位阶跃, } \frac{1}{s} \\ \text{输入单位斜坡, } \frac{1}{s^2} \\ \text{输入单位加速度, } \frac{1}{s^3} \end{cases}$

若输入信号是 $r(t)$ ，则 $R(s) = r(t)$ 结果里的 $\begin{cases} \text{常数项 变成 } \frac{\text{常数}}{s} \\ \text{常数} \cdot t \text{ 项 变成 } \frac{\text{常数}}{s^2} \\ \text{常数} \cdot t^2 \text{ 项 变成 } \frac{\text{常数}}{s^3} \end{cases}$

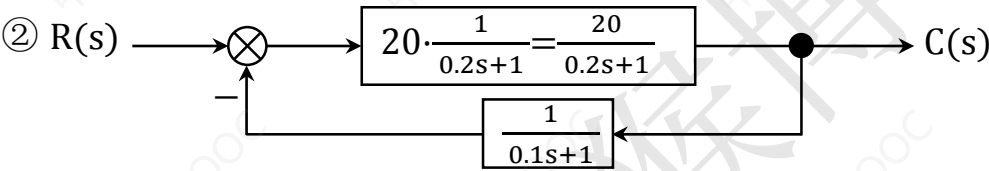
若输入信号是 $R(s)$ ，则 $R(s)$ 保持不变

稳态误差 $e_{ss}(\infty) = \lim_{s \rightarrow 0} s \cdot \left[1 - \frac{C(s)}{R(s)} H(s) \right] \cdot R(s)$

例2. 如图是某系统的结构图，设输入信号 $r(t) = 2 + 3t$ ，若系统稳定，用终值定理法求输入信号作用下的稳态误差 $e_{ss}(\infty)$



① $\frac{C(s)}{R(s)} = \frac{20(0.1s+1)}{(0.2s+1)(0.1s+1)+20}$ [第一章第3课学过]



$H(s) = \frac{1}{0.1s+1}$

③ $R(s) = \frac{2}{s} + \frac{3}{s^2}$

④
$$\begin{aligned} e_{ss}(\infty) &= \lim_{s \rightarrow 0} s \cdot \left[1 - \frac{C(s)}{R(s)} H(s) \right] \cdot R(s) \\ &= \lim_{s \rightarrow 0} s \cdot \left[1 - \frac{20(0.1s+1)}{(0.2s+1)(0.1s+1)+20} \times \frac{1}{0.1s+1} \right] \cdot \left(\frac{2}{s} + \frac{3}{s^2} \right) \\ &= \lim_{s \rightarrow 0} \frac{(0.2s+1)(0.1s+1)(2s+3)}{s[(0.2s+1)(0.1s+1)+20]} = \infty \quad \text{接近} \frac{(0.2 \times 0 + 1)(0.1 \times 0 + 1)(2 \times 0 + 3)}{0 \times [(0.2 \times 0 + 1)(0.1 \times 0 + 1) + 20]} = \frac{3}{0} = \infty \quad \text{【求极限时, } \frac{\text{非零数}}{0} = \infty \text{】} \end{aligned}$$

求开环零点与开环极点

例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$ ，试求出该系统的开环零点和开环极点，并将开环零点和开环极点画到坐标系上

① $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$

② 分子： $K^*(s+2) = 0 \Rightarrow s+2=0 \Rightarrow s_1=-2 \Rightarrow z_1=s_1=-2=-2+0\cdot j$

$$K^*(s+2) = 0 \Rightarrow \begin{cases} K^* = 0 \\ s+2 = 0 \end{cases}$$

$\therefore$ 求的是解 $s_1、s_2、s_3 \cdots$

$\therefore$ 只保留 $s+2=0$

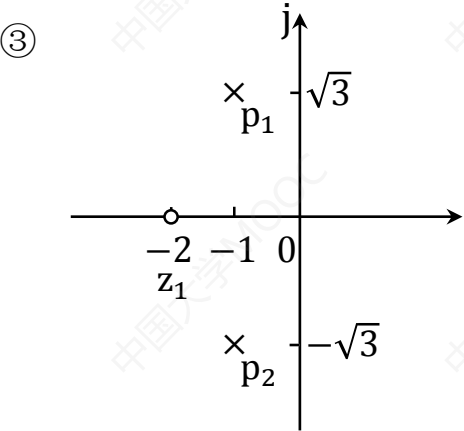
分母： $s^2+2s+4=0$

$$\Rightarrow \begin{cases} s_1 = \frac{-2+\sqrt{2^2-4\times4}}{2} \\ s_2 = \frac{-2-\sqrt{2^2-4\times4}}{2} \end{cases} \Rightarrow \begin{cases} s_1 = \frac{-2+\sqrt{-12}}{2} \\ s_2 = \frac{-2-\sqrt{-12}}{2} \end{cases} \Rightarrow \begin{cases} s_1 = \frac{-2+\sqrt{12}j}{2} \\ s_2 = \frac{-2-\sqrt{12}j}{2} \end{cases} \Rightarrow \begin{cases} s_1 = -1+\sqrt{3}j \\ s_2 = -1-\sqrt{3}j \end{cases}$$

$\Rightarrow p_1=s_1=-1+\sqrt{3}j, p_2=s_2=-1-\sqrt{3}j$

$$s^2+bs+c=0 \text{ 的解为 } \begin{cases} s_1 = \frac{-b+\sqrt{b^2-4c}}{2} \\ s_2 = \frac{-b-\sqrt{b^2-4c}}{2} \end{cases}$$

$\sqrt{-A} = \sqrt{A}j$



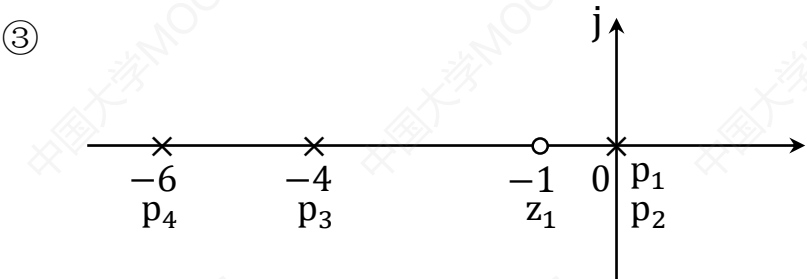
例2. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+1)}{s^2(s+4)(s+6)}$ ，试求出该系统的开环零点和开环极点，并将开环零点和开环极点画到坐标系上

① $G(s)H(s) = \frac{K^*(s+1)}{s^2(s+4)(s+6)}$

② 分子： $K^*(s+1) = 0 \Rightarrow s+1=0 \Rightarrow s_1=-1 \Rightarrow z_1=s_1=-1=-1+0\cdot j$

分母： $s^2(s+4)(s+6)=0$

$$\Rightarrow s\cdot s\cdot (s+4)(s+6) = 0 \Rightarrow \begin{cases} s_1 = 0 \\ s_2 = 0 \\ s_3 = -4 \\ s_4 = -6 \end{cases}$$
$$\Rightarrow \begin{cases} p_1 = s_1 = 0 = 0+0\cdot j \\ p_2 = s_2 = 0 \\ p_3 = s_3 = -4 = -4+0\cdot j \\ p_4 = s_4 = -6 = -6+0\cdot j \end{cases}$$



求开环零点与开环极点

例3. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)^3}{(s+1)^3}$ ，试求

出该系统的开环零点和开环极点，并将开环零点和

开环极点画到坐标系上

① $G(s)H(s) = \frac{K^*(s+2)^3}{(s+1)^3}$

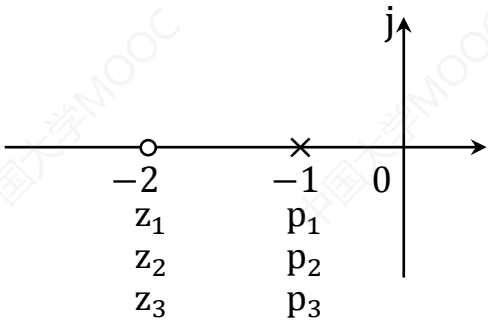
② 分子： $K^*(s+2)^3 = 0 \Rightarrow (s+2)^3 = 0 \Rightarrow (s+2) \cdot (s+2) \cdot (s+2) = 0$

$$\Rightarrow \begin{cases} s_1 = -2 \\ s_2 = -2 \\ s_3 = -2 \end{cases} \Rightarrow \begin{cases} z_1 = s_1 = -2 = -2 + 0 \cdot j \\ z_2 = s_2 = -2 = -2 + 0 \cdot j \\ z_3 = s_3 = -2 = -2 + 0 \cdot j \end{cases}$$

分母： $(s+1)^3 = 0$

$$\Rightarrow (s+1) \cdot (s+1) \cdot (s+1) = 0 \Rightarrow \begin{cases} s_1 = -1 \\ s_2 = -1 \\ s_3 = -1 \end{cases} \Rightarrow \begin{cases} p_1 = s_1 = -1 = -1 + 0 \cdot j \\ p_2 = s_2 = -1 = -1 + 0 \cdot j \\ p_3 = s_3 = -1 = -1 + 0 \cdot j \end{cases}$$

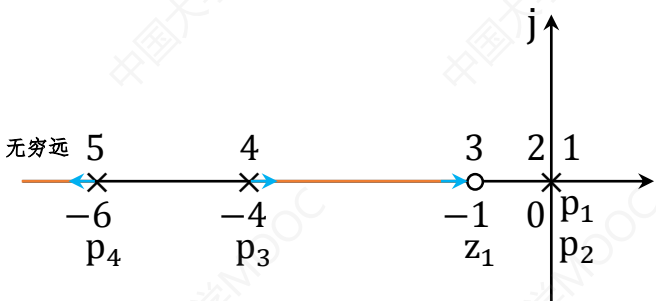
③



180° 根轨迹法—求分离点坐标

例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+1)}{s^2(s+4)(s+6)}$ ，求根轨迹
分离点坐标

$z_1 = -1$
 $p_1 = 0$ 、 $p_2 = 0$ 、 $p_3 = -4$ 、 $p_4 = -6$ 第三章第1课学过

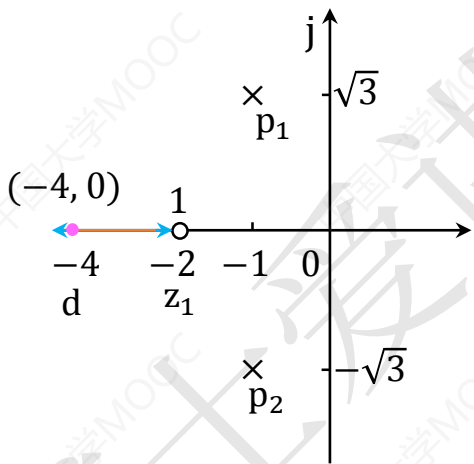


不存在分离点

例2. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$ ，求根轨迹
分离点坐标

$z_1 = -2$ 、 $p_1 = -1 + \sqrt{3}j$ 、 $p_2 = -1 - \sqrt{3}j$ 第三章第1课学过

$$\begin{aligned} \frac{1}{d-p_1} + \frac{1}{d-p_2} &= \frac{1}{d-z_1} \\ \Rightarrow \frac{1}{d-(-1+\sqrt{3}j)} + \frac{1}{d-(-1-\sqrt{3}j)} &= \frac{1}{d-(-2)} \\ \Rightarrow \frac{1}{(d+1)-\sqrt{3}j} + \frac{1}{(d+1)+\sqrt{3}j} &= \frac{1}{d+2} \\ \Rightarrow \frac{(d+1)+\sqrt{3}j+(d+1)-\sqrt{3}j}{[(d+1)-\sqrt{3}j][(d+1)+\sqrt{3}j]} &= \frac{1}{d+2} \\ \Rightarrow \frac{2d+2}{(d+1)^2-(\sqrt{3}j)^2} &= \frac{1}{d+2} \end{aligned}$$



$$\begin{aligned} \Rightarrow (2d+2)(d+2) &= (d+1)^2 - (\sqrt{3}j)^2 \\ \Rightarrow 2d^2 + 4d + 2d + 4 &= d^2 + 2d + 1 - (-3) \\ \Rightarrow d^2 + 4d &= 0 \\ \Rightarrow d(d+4) &= 0 \\ \Rightarrow d = 0 \text{ 或 } d = -4 \end{aligned}$$

根轨迹分离点坐标是 $(-4, 0)$

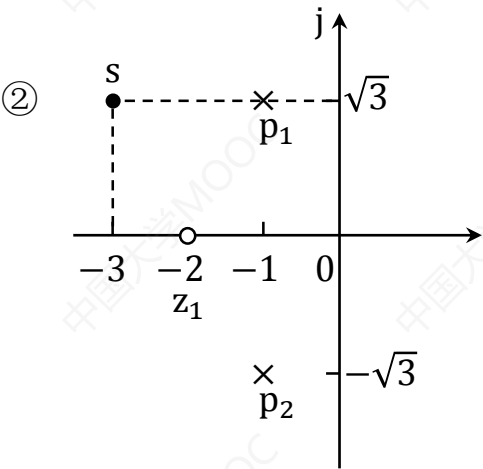
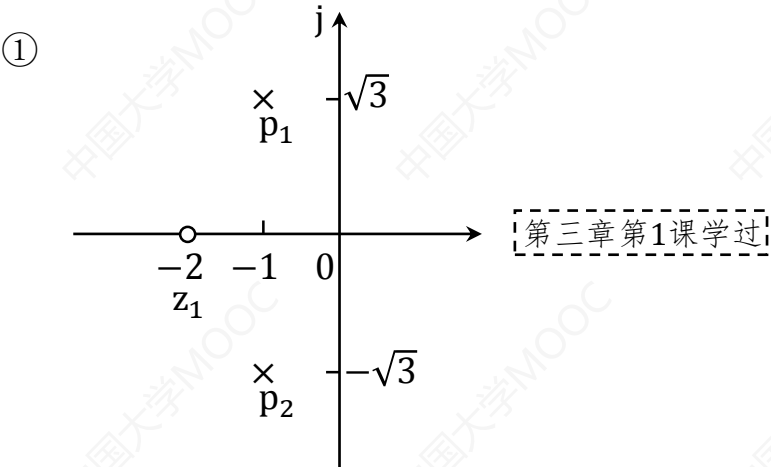
若 p 和 z 都求出：列 $\frac{1}{d-p_1} + \frac{1}{d-p_2} + \dots = \frac{1}{d-z_1} + \frac{1}{d-z_2} + \dots$ ，求 d

若只求出 p：列 $\frac{1}{d-p_1} + \frac{1}{d-p_2} + \dots = 0$ ，求 d

证明点是否在根轨迹上

例1. 某单位反馈系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$,

试证明点 $s = -3 + \sqrt{3}j$ 是否在根轨迹上



③ $\alpha = 120^\circ \times 1 = 120^\circ$

$\alpha_{\text{和}} = \alpha = 120^\circ$

④ $\beta = 180^\circ \times 1 = 180^\circ$

$\beta = 120^\circ \times 1 = 120^\circ$

$\beta_{\text{和}} = 180^\circ + 120^\circ = 300^\circ$ 奇数

⑤ $\alpha_{\text{和}} - \beta_{\text{和}} = 120^\circ - 300^\circ = -180^\circ = -1 \times 180^\circ$

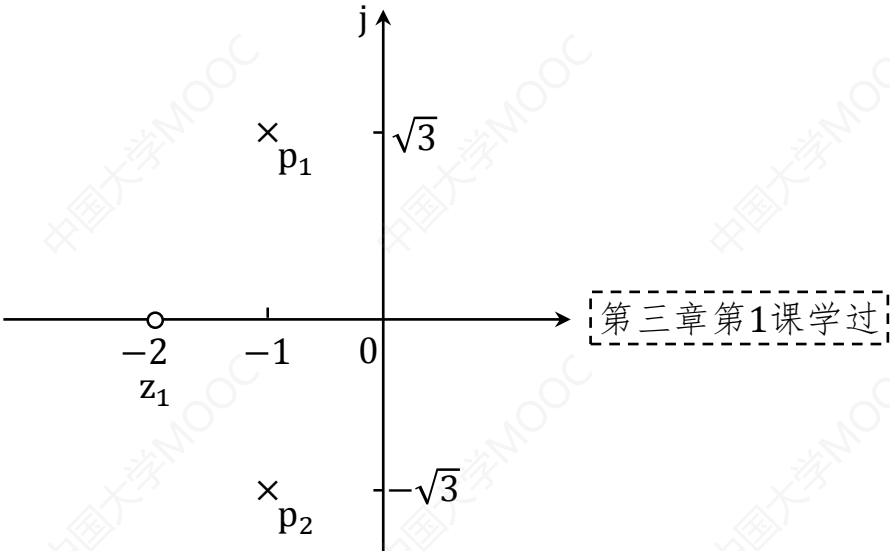
要证明的点在根轨迹上

180°根轨迹法—画起始角

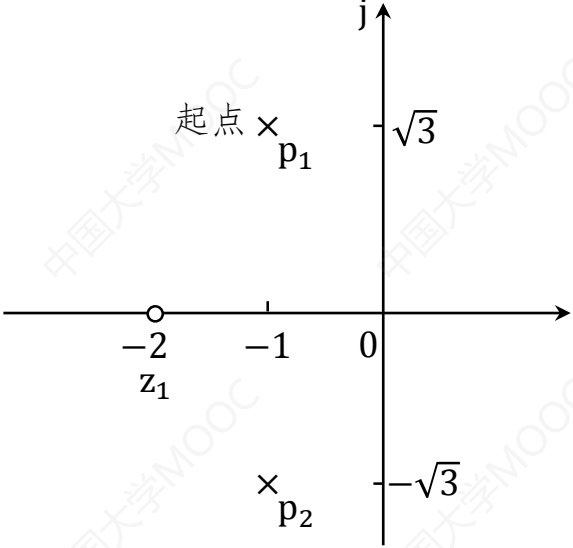
例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$,

画起始角

①



②

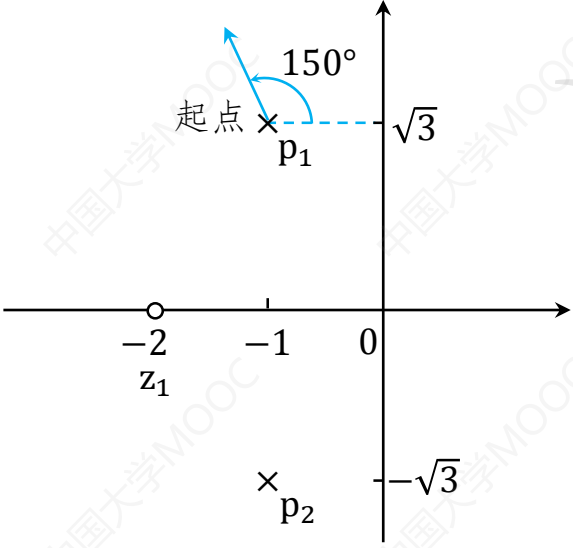


③ $\alpha_{\text{和}} = \alpha = 60^\circ$

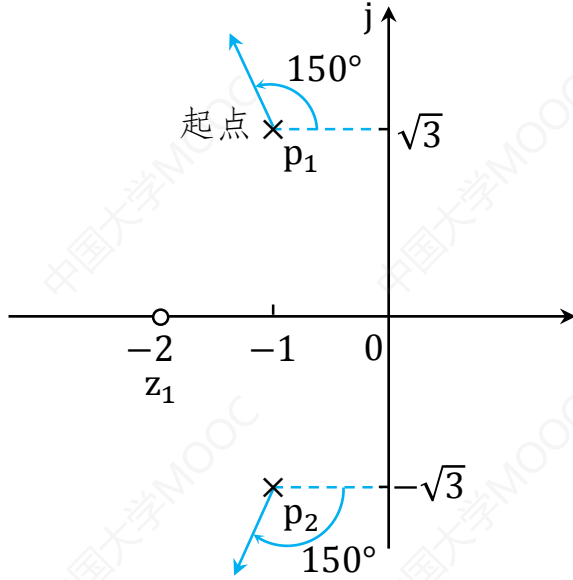
④ $\beta_{\text{和}} = \beta = 90^\circ$

⑤ $\theta_{\text{起点}} = \frac{180^\circ + \alpha_{\text{和}} - \beta_{\text{和}}}{\text{起点处 p 的个数}} = 150^\circ$

画 $\theta_{\text{起点}}$
起点



⑥

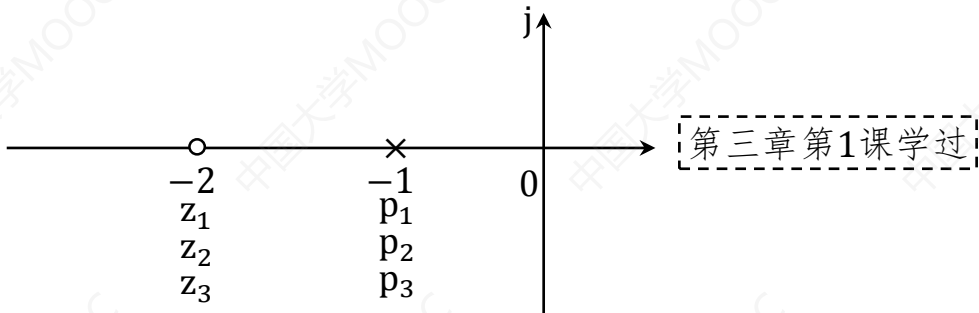


180°根轨迹法—画终止角

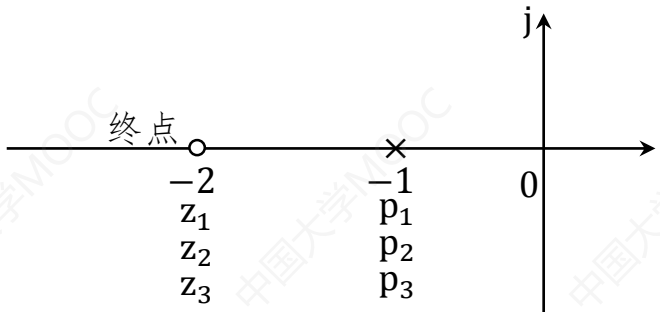
例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)^3}{(s+1)^3}$,

画终止角

①

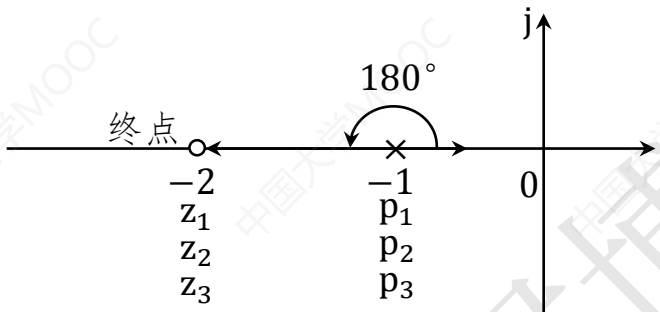


②



③ $\alpha_{\text{和}} = 0$

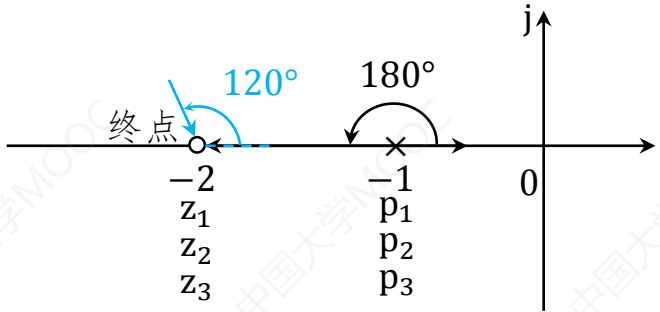
④



$$\beta = 180^\circ \times 3 = 540^\circ$$

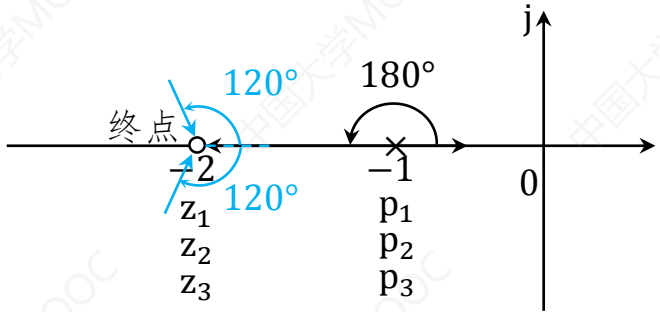
$$\beta_{\text{和}} = \beta = 540^\circ$$

⑤



$$\theta_{\text{终点}} = \frac{-180^\circ + \beta_{\text{和}} - \alpha_{\text{和}}}{\text{终点处 } z \text{ 的个数}} = \frac{-180^\circ + 540^\circ - 0}{3} = 120^\circ$$

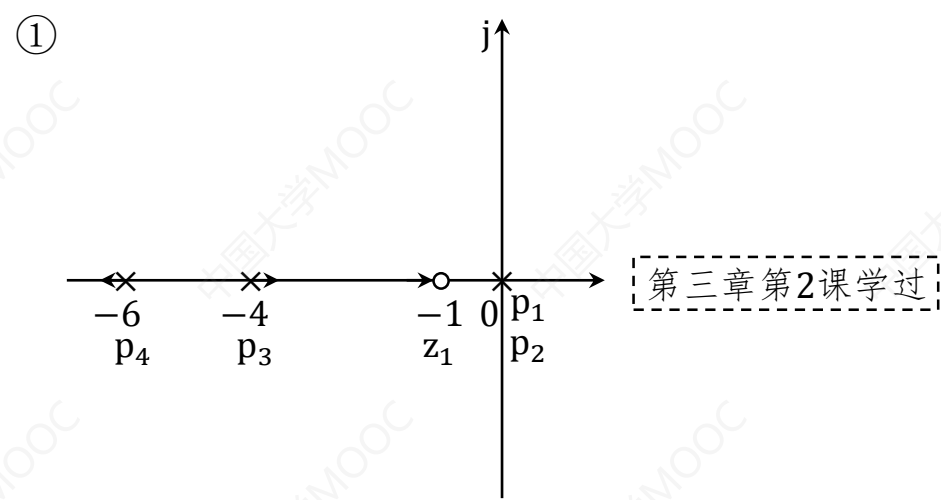
⑥



180° 根轨迹法—画根轨迹图

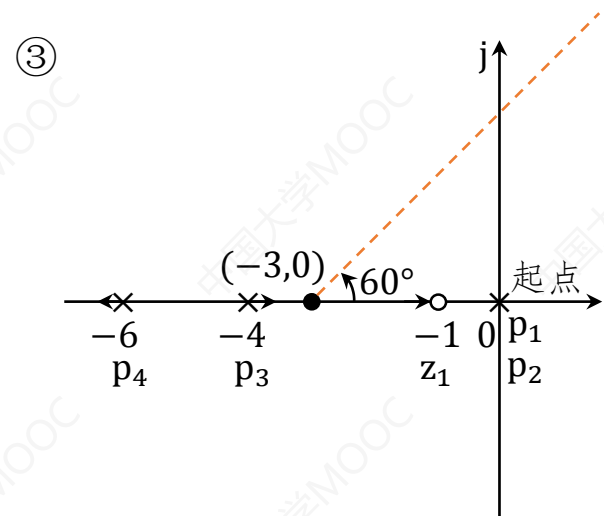
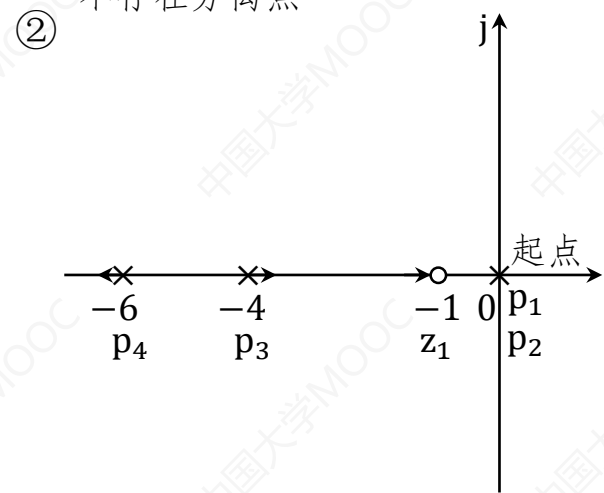
例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+1)}{s^2(s+4)(s+6)}$,

画系统的概略根轨迹图



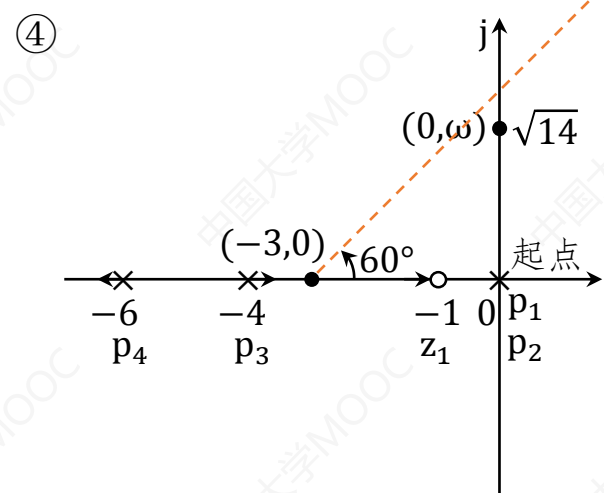
$z_1 = -1$
 $p_1 = 0 \quad p_2 = 0$
 $p_3 = -4 \quad p_4 = -6$

不存在分离点



$\varphi_a = \frac{\text{找到的奇数} \times 180^\circ}{|\text{p的个数} - \text{z的个数}|} = \frac{1 \times 180^\circ}{|4 - 1|} = 60^\circ$

点(-3,0)



$G(j\omega)H(j\omega) = \frac{K^*(j\omega+1)}{(j\omega)^2(j\omega+4)(j\omega+6)}$

$(j\omega)^2(j\omega+4)(j\omega+6) + K^*(j\omega+1) = 0$

$\Rightarrow \omega^4 - 10\omega^3j - 24\omega^2 + K^*j\omega + K^* = 0$

$\Rightarrow (\omega^4 - 24\omega^2 + K^*) + (K^*\omega - 10\omega^3)j = 0$

$a + bj = 0 \Rightarrow \begin{cases} a = 0 \\ b = 0 \end{cases}$

$\Rightarrow \begin{cases} \omega^4 - 24\omega^2 + K^* = 0 \\ K^*\omega - 10\omega^3 = 0 \end{cases} \Rightarrow \begin{cases} (\omega^2)^2 - 24\omega^2 + K^* = 0 \\ \omega(K^* - 10\omega^2) = 0 \end{cases}$

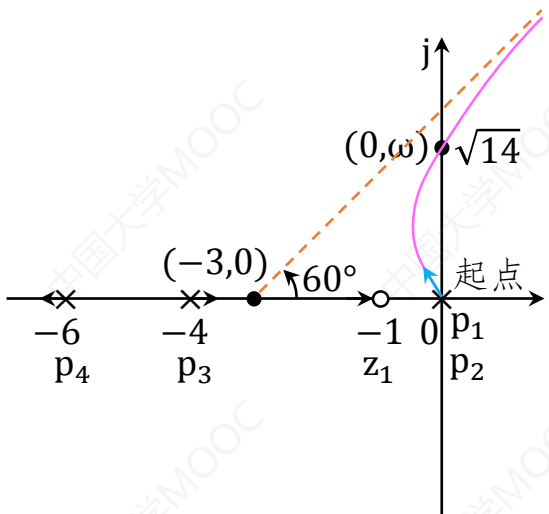
$\Rightarrow \begin{cases} \omega = \sqrt{14} \\ K^* = 140 \end{cases}$

由式子②可知

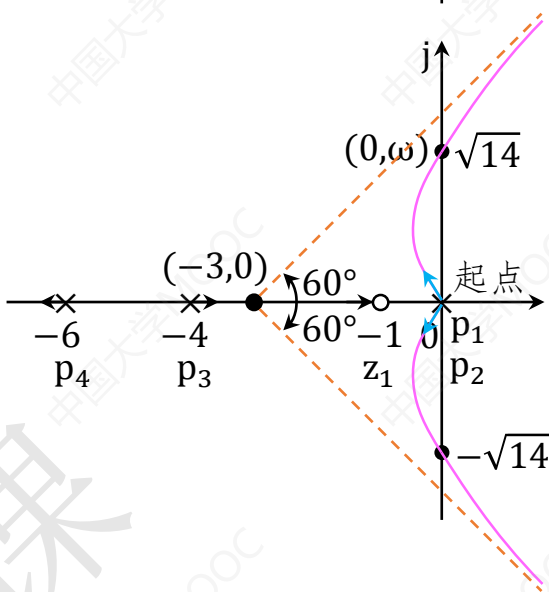
情况一: $\omega = 0$ 【将其代入式子①, 可求得 $K^*=0$ 】

情况二: $(K^* - 10\omega^2) = 0$ 即 $\omega^2 = \frac{K^*}{10}$ 【将其代入式子①, 可求得 $\begin{cases} \omega^2 = 14 \\ K^* = 140 \end{cases}$ 】

⑤

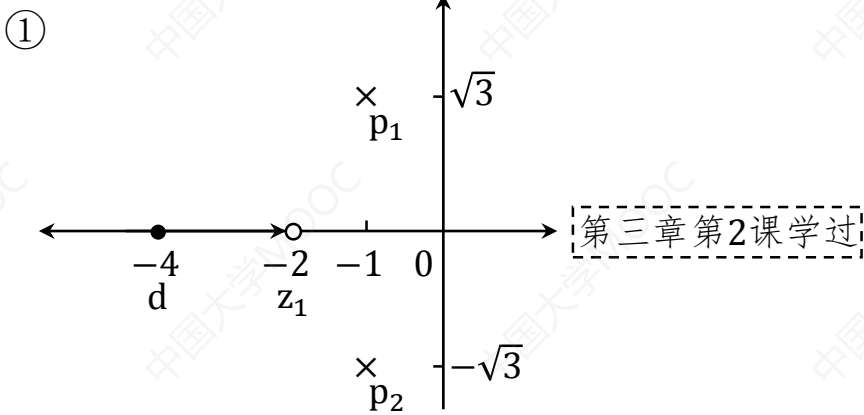


⑥

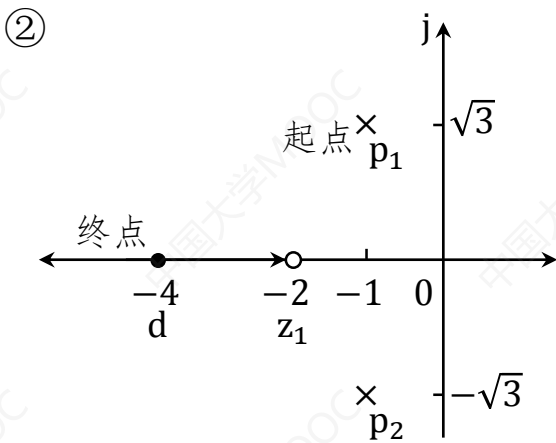


例2. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*(s+2)}{s^2+2s+4}$,

画系统的概略根轨迹图



$z_1 = -2$
 $p_1 = -1 + \sqrt{3}j$
 $p_2 = -1 - \sqrt{3}j$
根轨迹分离点坐标 $(-4, 0)$



③ 找不到

④ $G(j\omega)H(j\omega) = \frac{K^*(j\omega+2)}{(j\omega)^2+2j\omega+4}$

$(j\omega)^2 + 2j\omega + 4 + K^*(j\omega + 2) = 0$

$\Rightarrow -\omega^2 + 2j\omega + 4 + K^*j\omega + 2K^* = 0$

$\Rightarrow 2K^* - \omega^2 + 4 + (K^*\omega + 2\omega)j = 0$

$\Rightarrow \begin{cases} 2K^* - \omega^2 + 4 = 0 \\ K^*\omega + 2\omega = 0 \end{cases} \Rightarrow \begin{cases} 2K^* - \omega^2 + 4 = 0 & \text{①} \\ \omega(K^* + 2) = 0 & \text{②} \end{cases}$

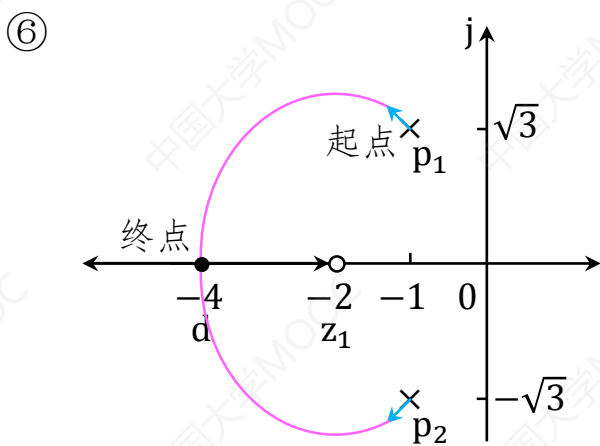
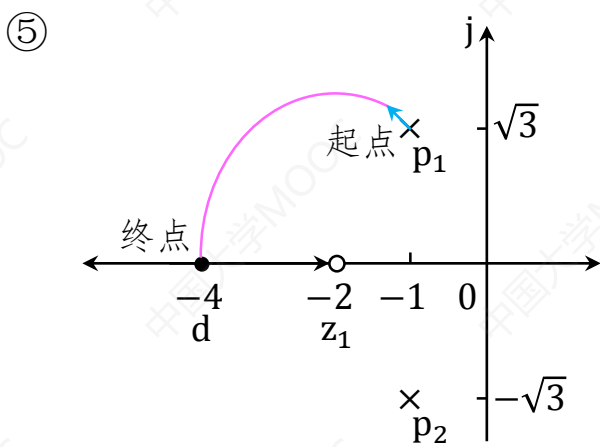
$\Rightarrow \begin{cases} \omega = 0 \\ K^* = -2 \end{cases}$

没有 ω 的解

由式子②可知

情况一： $\omega = 0$
【将其代入式子①，可求得 $K^* = -2$ 】

情况二： $K^* + 2 = 0$ 即 $K^* = -2$
【将其代入式子①，可求得 $\omega = 0$ 】



求根轨迹增益K*与开环增益K

例1. 某系统的开环传递函数为 $G(s)H(s) = \frac{K^*}{(s+1)(s+2)(s+4)}$,

求点 $s_1=-1+\sqrt{3}j$ 处的根轨迹增益 K^* 和开环增益 K

① $G(s)H(s) = \frac{K^*}{(s+1)(s+2)(s+4)}$

② $\left| \frac{K^*}{(-1+\sqrt{3}j+1)(-1+\sqrt{3}j+2)(-1+\sqrt{3}j+4)} \right| = 1$

$\Rightarrow \left| \frac{K^*}{\sqrt{3}j \cdot (1+\sqrt{3}j) \cdot (3+\sqrt{3}j)} \right| = 1$

整体的绝对值可以移动到每一个个体上

$\Rightarrow \frac{|K^*|}{|\sqrt{3}j \cdot (1+\sqrt{3}j) \cdot (3+\sqrt{3}j)|} = 1$

$\Rightarrow \frac{|K^*|}{|\sqrt{3}j| \cdot |1+\sqrt{3}j| \cdot |3+\sqrt{3}j|} = 1$

$\Rightarrow \frac{|K^*|}{|0+\sqrt{3}j| \cdot |1+\sqrt{3}j| \cdot |3+\sqrt{3}j|} = 1$

$|a+bj| = \sqrt{a^2 + b^2}$

$\Rightarrow \frac{|K^*|}{\sqrt{0^2+\sqrt{3}^2} \cdot \sqrt{1^2+\sqrt{3}^2} \cdot \sqrt{3^2+\sqrt{3}^2}} = 1$

$\Rightarrow |K^*| = 12$

$\Rightarrow K^* = 12$

④ $K = \frac{G(s)H(s) \text{ 分子里不含s的项}}{G(s)H(s) \text{ 分母里不含s的项}} = \frac{K^*}{8} = \frac{3}{2}$

例2. 某系统的开环传递函数为 $G(s)H(s) = \frac{K}{(s+1)(\frac{1}{2}s+1)(\frac{1}{4}s+1)}$,

求点 $s_1=-1+\sqrt{3}j$ 处的根轨迹增益 K^* 和开环增益 K

① $G(s)H(s) = \frac{K}{(s+1)(\frac{1}{2}s+1)(\frac{1}{4}s+1)} \stackrel{③}{=} \frac{K}{\frac{1}{8}s^3+\frac{7}{8}s^2+\frac{7}{4}s+1}$

② $\left| \frac{K}{(-1+\sqrt{3}j+1)[\frac{1}{2} \times (-1+\sqrt{3}j)+1][\frac{1}{4} \times (-1+\sqrt{3}j)+1]} \right| = 1$

$\Rightarrow \left| \frac{K}{\sqrt{3}j \cdot (\frac{1}{2}+\frac{\sqrt{3}}{2}j) \cdot (\frac{3}{4}+\frac{\sqrt{3}}{4}j)} \right| = 1$

$\Rightarrow \frac{|K|}{|\sqrt{3}j \cdot (\frac{1}{2}+\frac{\sqrt{3}}{2}j) \cdot (\frac{3}{4}+\frac{\sqrt{3}}{4}j)|} = 1$

$\Rightarrow \frac{|K|}{|\sqrt{3}j| \cdot |\frac{1}{2}+\frac{\sqrt{3}}{2}j| \cdot |\frac{3}{4}+\frac{\sqrt{3}}{4}j|} = 1$

$\Rightarrow \frac{|K|}{|0+\sqrt{3}j| \cdot |\frac{1}{2}+\frac{\sqrt{3}}{2}j| \cdot |\frac{3}{4}+\frac{\sqrt{3}}{4}j|} = 1$

$\Rightarrow \frac{|K|}{\sqrt{0^2+\sqrt{3}^2} \cdot \sqrt{(\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2} \cdot \sqrt{(\frac{3}{4})^2+(\frac{\sqrt{3}}{4})^2}} = 1$

$\Rightarrow |K| = \frac{3}{2}$

$\Rightarrow K = \frac{3}{2}$

④ $K^* = \frac{G(s)H(s) \text{ 分子里s最高的系数}}{G(s)H(s) \text{ 分母里s最高的系数}} = \frac{K}{\frac{1}{8}} = 12$

求开环频率特性

例1. 某单位反馈系统的开环传递函数为

G(s)H(s) = 300 / (s(s+10)(0.2s+1))

求系统的开环幅频特性和开环相频特性

① G(s)H(s) = 300 / (s(s+10)(0.2s+1))
G(jw)H(jw) = 300 / ((jw)^1(1+jw+10)^1(0.2jw+1)^1)
1 x 90° + 1 x arctan(1w/10) + 1 x arctan(0.2w/1)

- 原则一：不含j的项的相位角为0
- 原则二：(jw)^n的相位角为n*90°
- 原则三：(a*jw+b)^n的相位角为n*arctan(a/b)
- 原则四：若分子/分母=项1*项2*...
则分子/分母的相位角=项1的相位角+项2的相位角+...

② A(w) = |300 / (jw(jw+10)(0.2jw+1))|
= 300 / (|jw|*|jw+10|*|0.2jw+1|)
= 300 / (w*sqrt(100+w^2)*sqrt(1+0.04w^2))

φ(w) = 0 - (1 x 90° + 1 x arctan(1w/10) + 1 x arctan(0.2w/1)) = -90° - arctan(0.1w) - arctan(0.2w)

φ(w) = G(jw)H(jw)分子相位角 - G(jw)H(jw)分母相位角

例2. 某单位反馈系统的开环传递函数为

G(s)H(s) = (s+1)^2 / s^3

求系统的开环

① G(s)H(s) = (s+1)^2 / s^3
G(jw)H(jw) = (1+jw+1)^2 / (jw)^3
2 x arctan(1w/1)
3 x 90°

② A(w) = |(jw+1)^2 / (jw)^3|
= (|jw+1|*|jw+1|) / (|jw|*|jw|*|jw|)
= (sqrt(1+w^2)*sqrt(1+w^2)) / (w*w*w)
= (1+w^2) / w^3

φ(w) = 2*arctan(1w/1) - 3*90° = 2arctan w - 270°

求闭环频率特性和输出信号

例1. 某单位反馈系统的闭环传递函数为 $\frac{C(s)}{R(s)} = \frac{1}{s+2}$ ，
输入信号为 $r(t) = 3\sin(2t + 30^\circ)$ ，求系统的
闭环幅频特性、闭环相频特性以及输出信号

例2. 某单位反馈系统的闭环传递函数为 $\frac{C(s)}{R(s)} = \frac{1}{s+2}$ ，
输入信号为 $r(t) = 3\sin(2t + 30^\circ)$ ，求系统的
闭环频率特性以及输出信号

闭环幅频特性 闭环相频特性

① $\frac{C(s)}{R(s)} = \frac{1}{s+2}$
 $\Downarrow$
 $\frac{C(j\omega)}{R(j\omega)} = \frac{1}{(1 \cdot j\omega + 2)^1}$

$\nearrow 0$
 $\searrow 1 \cdot \arctan \frac{1 \cdot \omega}{2}$

原则一：不含 j 的项的相位角为 0
原则二：(jω)ⁿ 的相位角为 n·90°
原则三：(a·jω + b)ⁿ 的相位角为 n·arctan $\frac{a\omega}{b}$
原则四：若 分子/分母=项1·项2·...
则 分子/分母的相位角=项1的相位角+项2的相位角+ ...

② $A(\omega) = \left| \frac{1}{j\omega+2} \right|$
 $= \frac{|1|}{|j\omega+2|}$
 $= \frac{1}{\sqrt{2^2+\omega^2}}$
 $= \frac{1}{\sqrt{4+\omega^2}}$

闭环幅频特性： $A(\omega) = \left| \frac{C(j\omega)}{R(j\omega)} \right|$

$\varphi(\omega) = 0 - 1 \cdot \arctan \frac{1 \cdot \omega}{2} = -\arctan \frac{\omega}{2}$

闭环相频特性： $\varphi(\omega) = \frac{C(j\omega)}{R(j\omega)}$ 分子相位角 - $\frac{C(j\omega)}{R(j\omega)}$ 分母相位角

③ $a = 3$ 、 $\omega_0 = 2$ 、 $\alpha = 30^\circ$

$A(\omega_0) = \frac{1}{\sqrt{4+\omega_0^2}}$ $\varphi(\omega_0) = -\arctan \frac{\omega_0}{2}$

输出信号： $c(t) = A(\omega_0) \cdot a \cdot \sin[\omega_0 t + \alpha + \varphi(\omega_0)]$

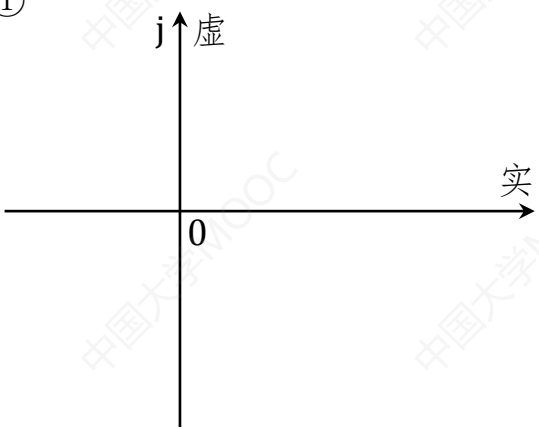
$c(t) = \frac{1}{\sqrt{4+2^2}} \cdot 3 \cdot \sin \left[2t + 30^\circ + \left(-\arctan \frac{2}{2} \right) \right] = \frac{3}{2\sqrt{2}} \cdot \sin [2t + 30^\circ + (-45^\circ)] = \frac{3}{2\sqrt{2}} \sin(2t - 15^\circ)$

画奈奎斯特曲线/开环幅相曲线

例1. 某单位反馈系统的开环传递函数为 $G(s)H(s) = \frac{(s+1)^2}{s^3}$,
请画出系统的奈奎斯特曲线

例2. 某单位反馈系统的开环传递函数为 $G(s)H(s) = \frac{(s+1)^2}{s^3}$,
请画出系统的开环幅相曲线

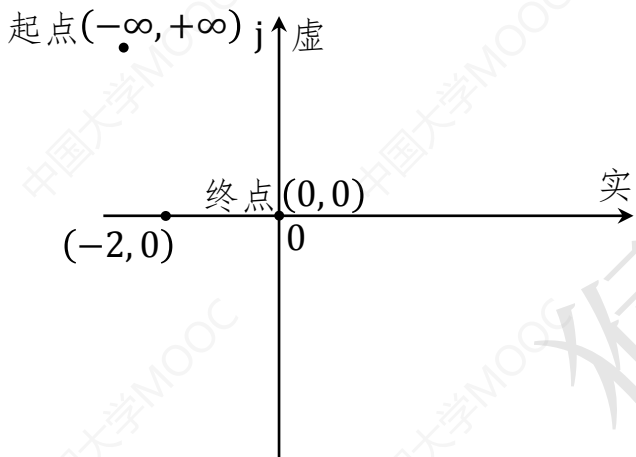
①



② $G(s)H(s) = \frac{(s+1)^2}{s^3}$

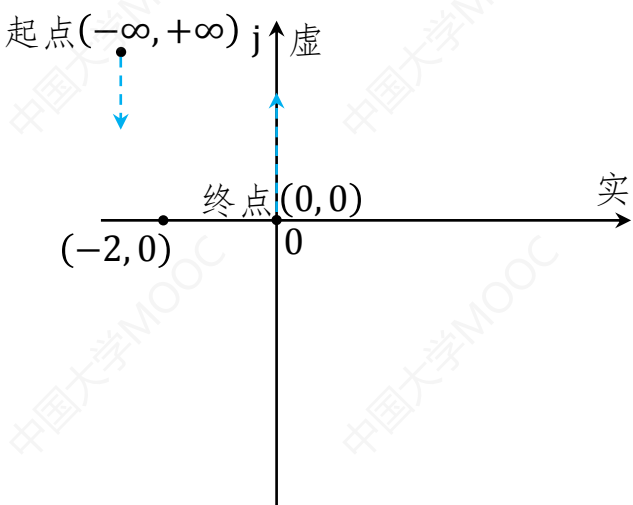
③ $G(j\omega)H(j\omega) = \frac{(j\omega+1)^2}{(j\omega)^3}$
 $= -\frac{2}{\omega^2} + \frac{1-\omega^2}{\omega^3}j$

(1) $-\frac{2}{0^2} + \frac{1-0^2}{0^3}j = -\infty + \infty \cdot j$
(2) $-\frac{2}{(+\infty)^2} + \frac{1-(+\infty)^2}{(+\infty)^3}j = 0 + 0 \cdot j$
(3) $\frac{1-\omega^2}{\omega^3} = 0 \Rightarrow 1 - \omega^2 = 0 \Rightarrow \omega = \pm 1$
 $-\frac{2}{(\pm 1)^2} + \frac{1-(\pm 1)^2}{(\pm 1)^3}j = -2 + 0 \cdot j$
(4) $-\frac{2}{\omega^2} = 0 \Rightarrow$ 无解

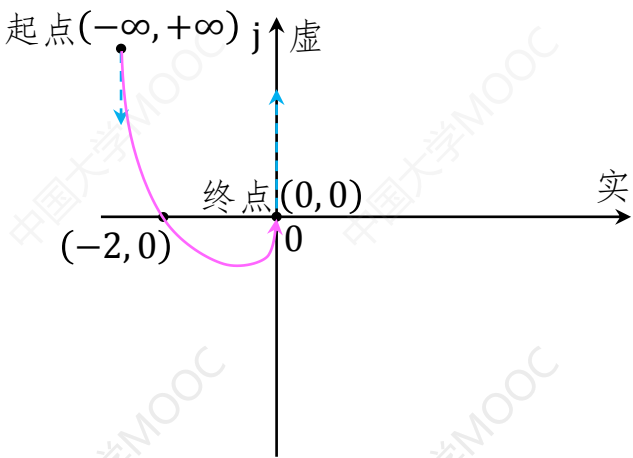


④ $\varphi(\omega) = 2\arctan \omega - 270^\circ$ 第四章第1课学过
 $\varphi(0) = 2\arctan 0 - 270^\circ = 2 \times 0 - 270^\circ = -270^\circ$
 $\varphi(+\infty) = 2\arctan(+\infty) - 270^\circ = 2 \times 90^\circ - 270^\circ = -90^\circ$

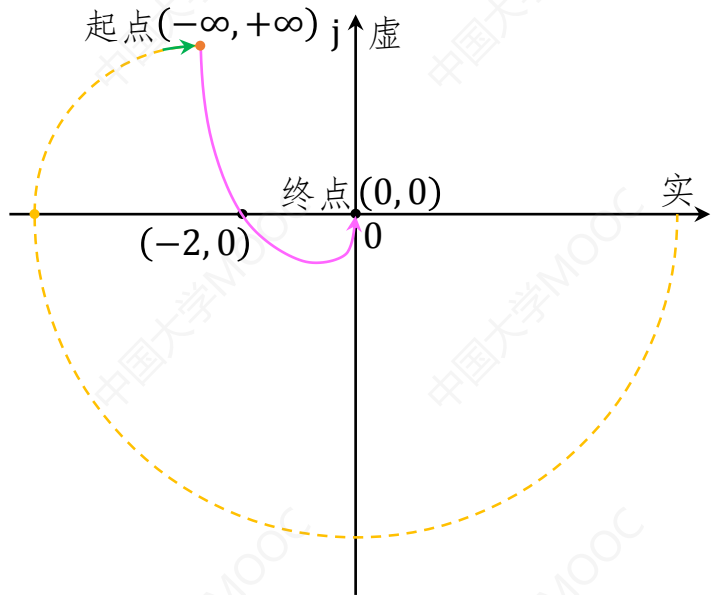
⑤



⑥



⑥



奈奎斯特稳定判据

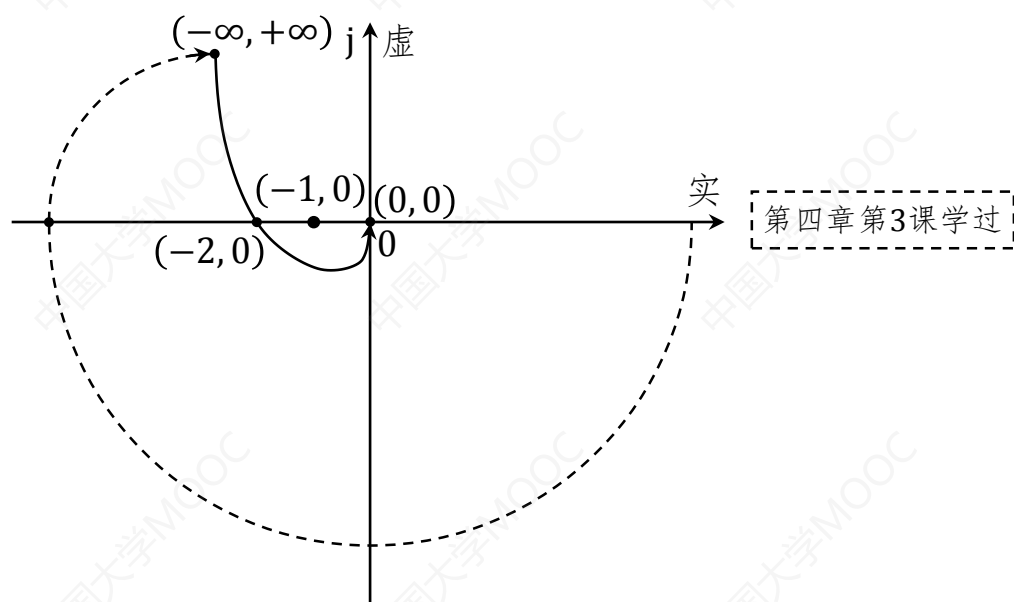
例1. 某单位反馈系统的开环传递函数 $G(s)H(s) = \frac{(s+1)^2}{s^3}$,

试通过奈奎斯特稳定判据判断系统的稳定性,

并判断 s 右半平面闭环极点的个数。

$$\textcircled{1} \quad G(s)H(s) = \frac{(s+1)^2}{s^3}$$

②



$$\textcircled{3} \quad s^3 = 0$$

$$\Rightarrow s \cdot s \cdot s = 0$$

$$\Rightarrow \begin{cases} s_1 = 0 \\ s_2 = 0 \\ s_3 = 0 \end{cases}$$

$$\Rightarrow P = 0$$

④ 演算纸:

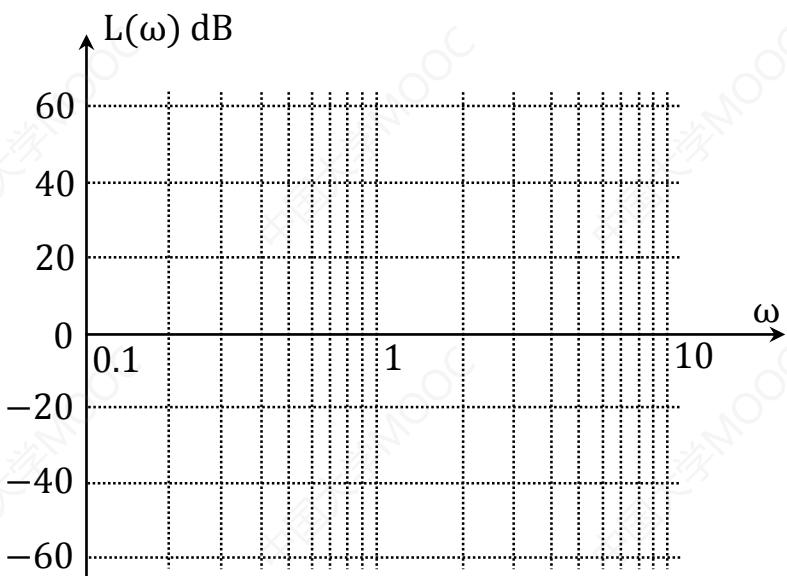
$$N = 0$$

$$\textcircled{5} \quad P - 2N = 0 - 2 \times 0 = 0$$

系统稳定, $\begin{cases} \text{正实部闭环极点个数} \\ \text{s右半平面闭环极点个数} \end{cases}$ 为 0

画开环对数幅频渐进特性曲线/伯德图

例1. 某单位反馈系统的开环传递函数为 $G(s)H(s) = \frac{300}{(s^2+10s)(0.2s+1)}$,
试画出系统开环对数幅频渐进特性曲线



①
$$G(s)H(s) = \frac{300}{(s^2+10s)(0.2s+1)}$$
$$= \frac{300 \times 1 \times 1}{10 \cdot s \cdot (0.1s+1) \cdot (0.2s+1)}$$
$$= 30 \cdot \frac{1}{s^1} \cdot \frac{1}{(0.1s+1) \cdot (0.2s+1)}$$

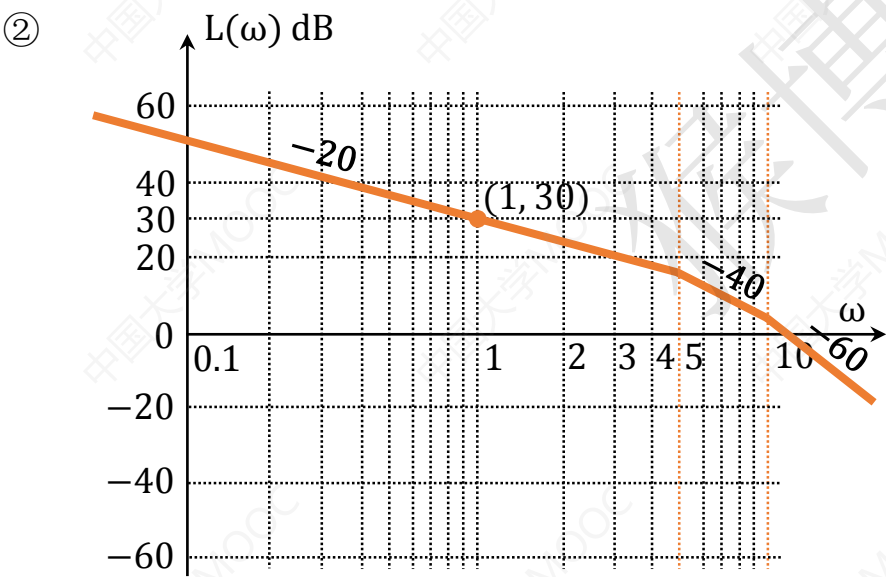
$$\omega = \frac{1}{0.1} = 10$$

斜率变化量: -20

$$\omega = \frac{1}{0.2} = 5$$

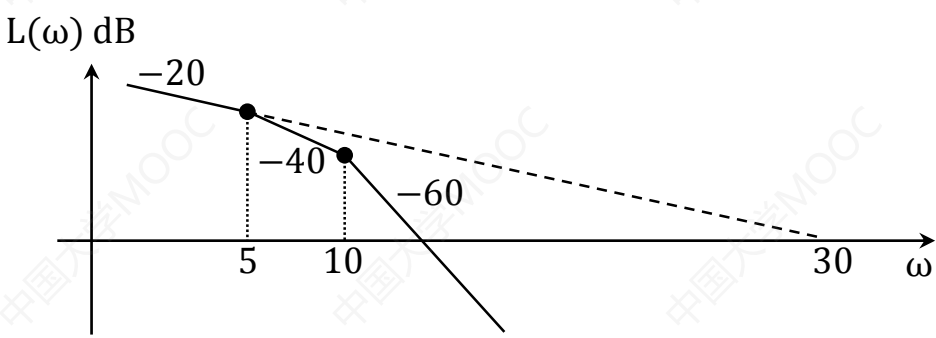
斜率变化量: -20

首段经过点: $(1, 20 \cdot \lg 30) \approx (1, 30)$
初始斜率: $-20 \times 1 = -20$



根据伯德图求开环传递函数

例1. 已知某系统的伯德图如下，求该系统的开环传递函数



① $v = \frac{-20}{-20} = 1$

$K = 30^1 = 30$

② 式子 $= \frac{1}{\frac{1}{5} \cdot s + 1} = \frac{1}{0.2s + 1}$

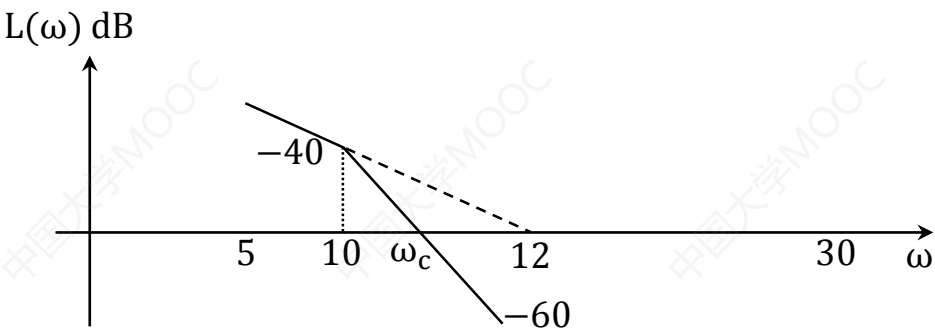
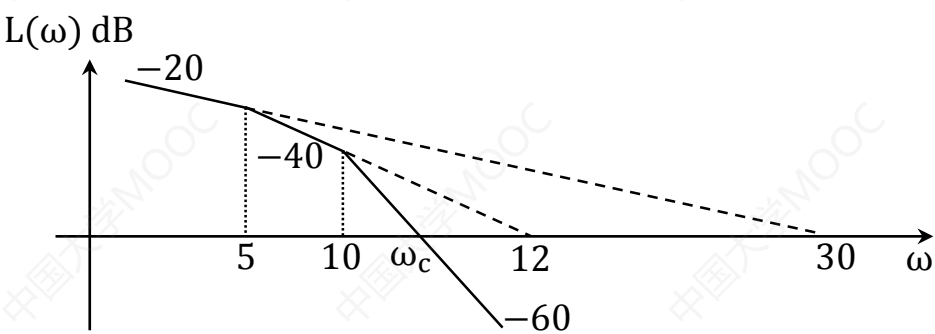
式子 $= \frac{1}{\frac{1}{10} \cdot s + 1} = \frac{1}{0.1s + 1}$

开环传递函数 $G(s)H(s) = 30 \cdot \frac{1}{s^1} \cdot \left(\frac{1}{0.2s + 1} \cdot \frac{1}{0.1s + 1} \right) = \frac{30}{s(0.2s + 1)(0.1s + 1)}$

开环传递函数 $G(s)H(s) = K \cdot \frac{1}{s^v} \cdot$ 每个转折点式子的乘积

根据伯德图求ω

例1. 已知某系统的伯德图如下，求 ω_c

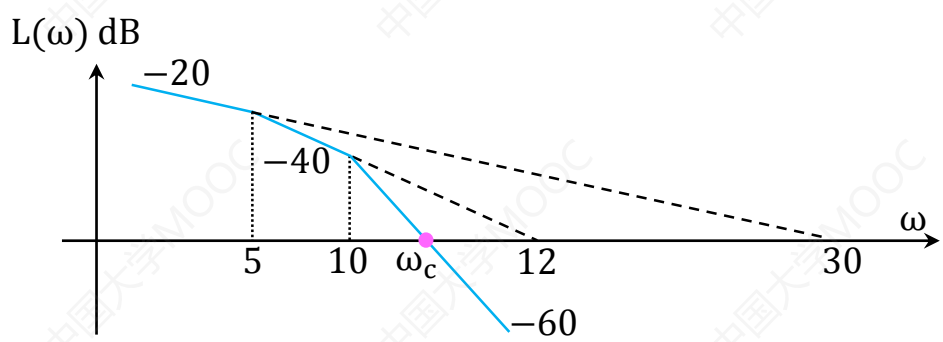


$$|-40| \cdot \lg \frac{12}{10} = |-60| \cdot \lg \frac{\omega_c}{10}$$
$$\Rightarrow 40 \cdot \lg \frac{12}{10} = 60 \cdot \lg \frac{\omega_c}{10}$$
$$\Rightarrow 2 \cdot \lg \frac{6}{5} = 3 \cdot \lg \frac{\omega_c}{10}$$
$$\Rightarrow \lg \left(\frac{6}{5}\right)^2 = \lg \left(\frac{\omega_c}{10}\right)^3$$
$$\Rightarrow \left(\frac{6}{5}\right)^2 = \left(\frac{\omega_c}{10}\right)^3$$
$$\Rightarrow \frac{36}{25} = \frac{\omega_c^3}{1000}$$
$$\Rightarrow \omega_c^3 = 1440$$
$$\Rightarrow \omega_c = 11.29 \text{ rad/s}$$

$$|\text{斜率1}| \cdot \lg \frac{\omega_\gamma}{\omega_\alpha} = |\text{斜率2}| \cdot \lg \frac{\omega_\beta}{\omega_\alpha}$$
$$a \cdot \lg b = \lg b^a$$
$$\lg a = \lg b \Rightarrow a = b$$

根据伯德图求截止频率 ω_c

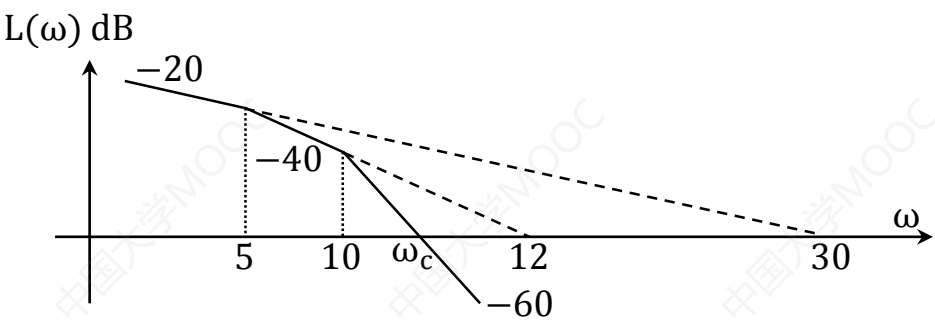
例1. 已知某系统的伯德图如下，求截止频率 ω_c



$\omega_c = 11.29 \text{ rad/s}$ 第四章第7课学过

求系统的稳定裕度

例1. 某单位反馈系统的开环传递函数 $G(s)H(s) = \frac{300}{s(s+10)(0.2s+1)}$ ，系统的伯德图如下，求该系统的相角裕度 γ 和幅值裕度 h



- ① $\varphi(\omega) = -90^\circ - \arctan(0.1\omega) - \arctan(0.2\omega)$ 第四章第1课学过
 $\omega_c = 11.29 \text{ rad/s}$ 第四章第8课学过
- ② $\gamma = 180^\circ + (-204.58^\circ) = -24.58^\circ$ 相角裕度 $\gamma = 180^\circ + \varphi(\omega_c)$
【 $\varphi(\omega_c) = -90^\circ - \arctan(0.1 \times 11.29) - \arctan(0.2 \times 11.29) = -204.58^\circ$ 】
- ③ $-90^\circ - \arctan(0.1\omega_g) - \arctan(0.2\omega_g) = -180^\circ$
【 $\varphi(\omega_g) = -90^\circ - \arctan(0.1\omega_g) - \arctan(0.2\omega_g)$ 】

$$\arctan(0.1\omega_g) + \arctan(0.2\omega_g) = 90^\circ$$
$$\frac{0.1\omega_g + 0.2\omega_g}{1 - 0.1\omega_g \cdot 0.2\omega_g} = \tan 90^\circ$$
$$\frac{0.3\omega_g}{1 - 0.02(\omega_g)^2} = +\infty$$
$$1 - 0.02(\omega_g)^2 = 0$$
$$\omega_g = \sqrt{50} \text{ rad/s}$$

$$\arctan a + \arctan b = c \Rightarrow \frac{a+b}{1-a \cdot b} = \tan c$$
$$\arctan a - \arctan b = c \Rightarrow \frac{a-b}{1+a \cdot b} = \tan c$$

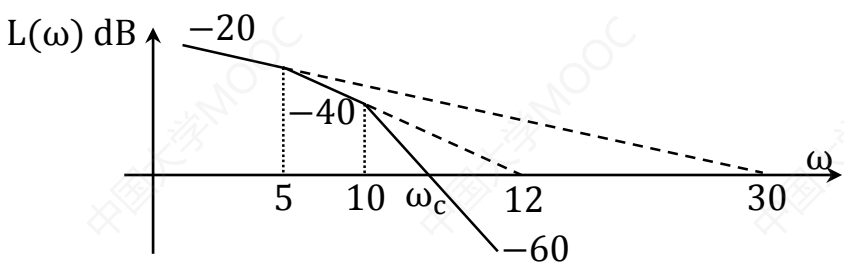
④ $G(j\omega_g)H(j\omega_g) = \frac{300}{j\omega_g(j\omega_g+10)(0.2j\omega_g+1)} = \frac{300}{j \times \sqrt{50} \times (j \times \sqrt{50} + 10)(0.2j \times \sqrt{50} + 1)}$

$$h = \frac{1}{\left| \frac{300}{j \times \sqrt{50} \times (j \times \sqrt{50} + 10)(0.2j \times \sqrt{50} + 1)} \right|} \text{ dB}$$
$$= \frac{1}{\frac{|300|}{|j \times \sqrt{50}| \cdot |j \times \sqrt{50} + 10| \cdot |0.2j \times \sqrt{50} + 1|}} \text{ dB}$$
$$= \frac{1}{\frac{300}{|0 + \sqrt{50}j| \cdot |10 + \sqrt{50}j| \cdot |1 + 0.2 \times \sqrt{50}j|}} \text{ dB}$$
$$= \frac{1}{\frac{300}{\sqrt{0^2 + \sqrt{50}^2} \cdot \sqrt{10^2 + \sqrt{50}^2} \cdot \sqrt{1^2 + (0.2 \times \sqrt{50})^2}}} \text{ dB}$$
$$= \frac{1}{2} \text{ dB}$$

幅值裕度 $h = \frac{1}{|G(j\omega_g)H(j\omega_g)|} \text{ dB}$

线性系统的串联校正

例1. 某系统开环传递函数 $G(s)H(s) = \frac{300}{s(s+10)(0.2s+1)}$ ，伯德图如下，要求校正后: 截止频率 $\geq 2.3 \text{ rad/s}$ ，相角裕度 $\geq 40^\circ$ ，判断该系统应该采用的校正方式，并写出校正函数以及校正后的开环传递函数。



① $G(s)H(s) = \frac{300}{s(s+10)(0.2s+1)}$ [第一章第3课和第四章第6课学过]

② $\omega_c = 11.29 \text{ rad/s}$ [第四章第8课学过]

③ $\gamma = -24.58^\circ$ [第四章第9课学过]

④ a、 $\omega_c^* = 2.3 \text{ rad/s}$

b、 $\gamma^* = 40^\circ$

c、 $\omega_c^* < \omega_c, \gamma^* > \gamma \Rightarrow$ 校正方式为串联滞后校正

⑤ $G(j\omega_c^*)H(j\omega_c^*) = \frac{300}{j\omega_c^*(j\omega_c^*+10)(0.2j\omega_c^*+1)} = \frac{300}{j \times 2.3 \times (j \times 2.3 + 10)(0.2j \times 2.3 + 1)}$

$$G_c(s) = \frac{\frac{1}{0.1 \times 2.3} \cdot s + 1}{\frac{\frac{1}{j \times 2.3 \times (j \times 2.3 + 10)(0.2j \times 2.3 + 1)}}{0.1 \times 2.3} \cdot s + 1} = \frac{\frac{1}{0.1 \times 2.3} \cdot s + 1}{\frac{11.55}{0.1 \times 2.3} \cdot s + 1} = \frac{4.35s + 1}{50.22s + 1}$$

校正后的开环传递函数 = $\frac{300}{s(s+10)(0.2s+1)} \cdot \frac{4.35s+1}{50.22s+1} = \frac{300(4.35s+1)}{s(s+10)(0.2s+1)(50.22s+1)}$

校正函数：

串联超前校正： $G_c(s) = \frac{\sqrt{\frac{1 + \sin(\gamma^* - \gamma + 7^\circ)}{1 - \sin(\gamma^* - \gamma + 7^\circ)}} \cdot \frac{1}{\omega_c^*} \cdot s + 1}{\frac{1}{\sqrt{\frac{1 + \sin(\gamma^* - \gamma + 7^\circ)}{1 - \sin(\gamma^* - \gamma + 7^\circ)}} \cdot \omega_c^*} \cdot s + 1}$

串联滞后校正：用 $j\omega_c^*$ 替换 $G(s)H(s)$ 中的 s ，得到 $G(j\omega_c^*)H(j\omega_c^*)$ ，

$$G_c(s) = \frac{\frac{1}{0.1 \cdot \omega_c^*} \cdot s + 1}{\frac{|G(j\omega_c^*)H(j\omega_c^*)|}{0.1 \cdot \omega_c^*} \cdot s + 1}$$

校正后的开环传递函数 = 原来的 $G(s)H(s) \cdot G_c(s)$